AUMOVIO

Intelligent Mutex Execution Environment

Complete Application User Guide

IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

AUMOVIO

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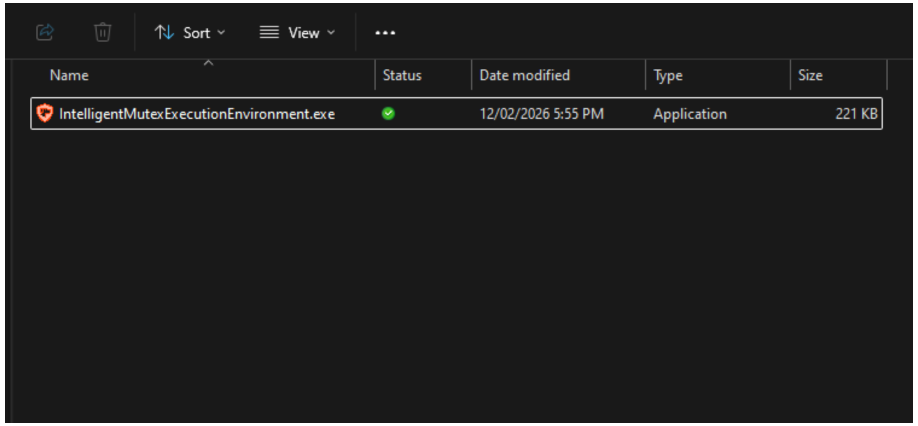
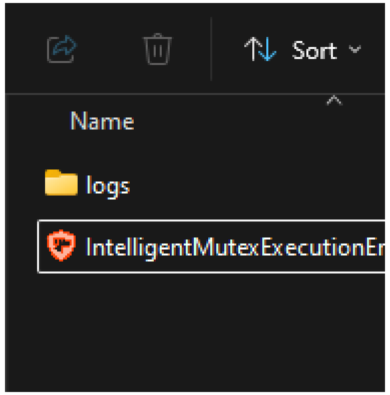
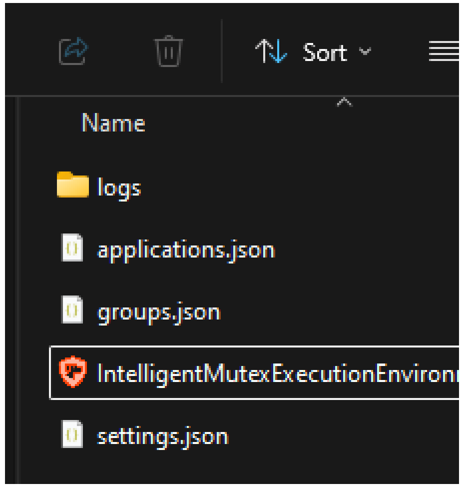
IMEE Installation



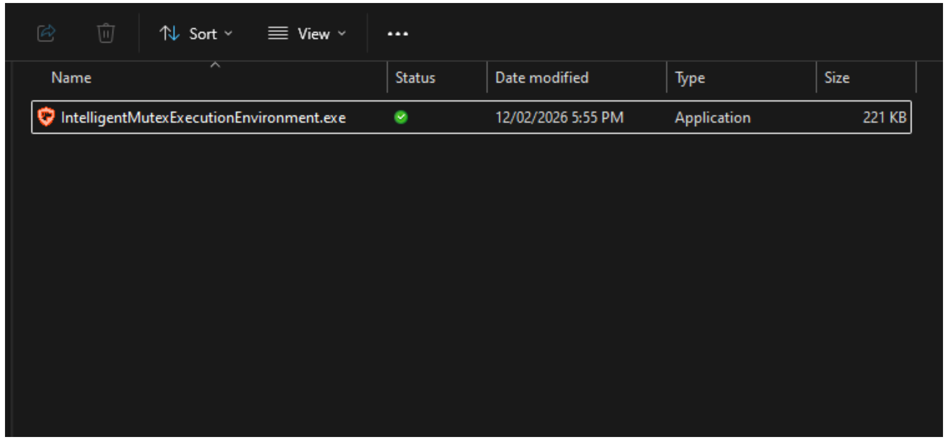


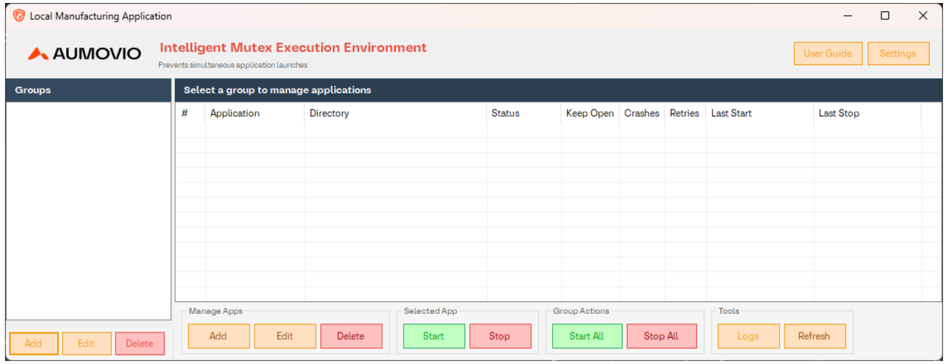


No installation is required. Copy \`IntelligentMutexExecutionEnvironment.exe\` to any folder and run it. The application creates its data files automatically in the same directory:

File	Purpose
application.json	Stores your managed applications
groups.json	Stores your groups
settings.json	Stores your station name, performance, and logging configuration
logs/	Daily log files

Getting Started
First launch and Startup Behavior





First launch

- Double-click `IntelligentMutexExecutionEnvironment.exe`
- The window appears in the **bottom-right corner** of your screen
  - Both the group list (left) and application list (right) will be empty

Startup Behavior

When Instance Manager starts, it checks if any managed applications are already running. If so, they are **automatically terminated** and a notification lists the affected apps. This ensures all managed applications are only launched through Intelligent Mutex Execution Environment.

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Interface Overview for IMEE

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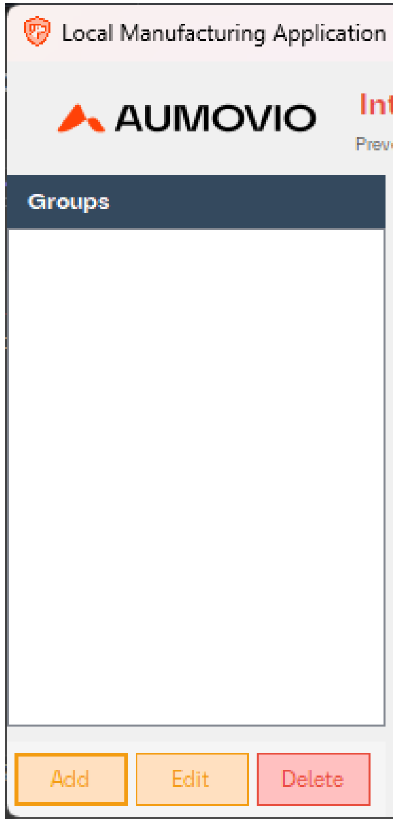
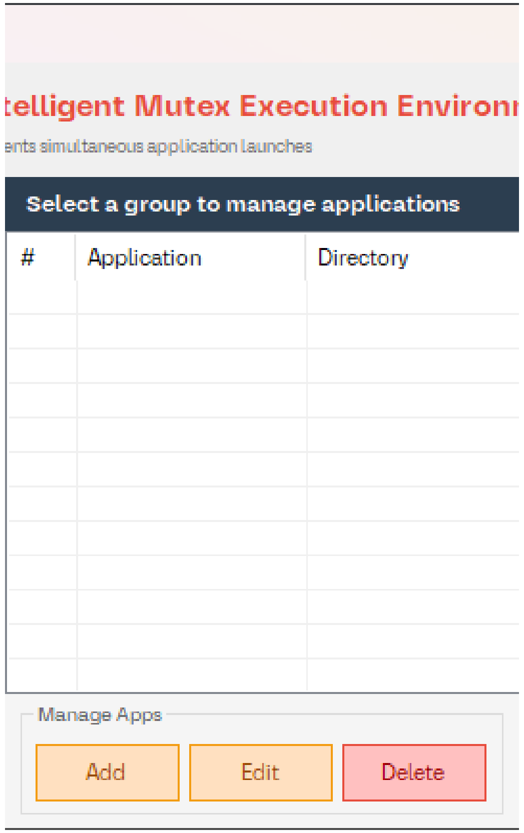
Interface Overview
The interface is divided into two main areas



Left Panel - Groups
Group list : Shows all your application groups
Add Group / Edit Group / Delete Group: To Manage groups

Column	Description
Indicator	Current state (see Group Status indicators)
Group Name	Name of the Executable
N of T	Current state (see Group Status indicators)

Interface Overview
The interface is divided into two main areas



Right Panel Applications
Application List: Shows apps in the selected group with 9 columns
Action Buttons: Add, Edit, Delete, Start, Stop, Start All, Stop All, Refresh
Settings / Logs: Configure station name, performance, and logging settings, or open the logs folder

Column	Description
Index	Unique auto-incremented ID
Application	Name of the Executable
Directory	Full file path
Status	Current state (see Status indicators)
Keep Open	“Yes” if auto-restart is enabled
Crashes	Total Number of detected crashes
Retries	Current auto-restart retry count
Last Start	When the app was last started
Last Stop	When the app was last stopped

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Managing Groups for IMEE

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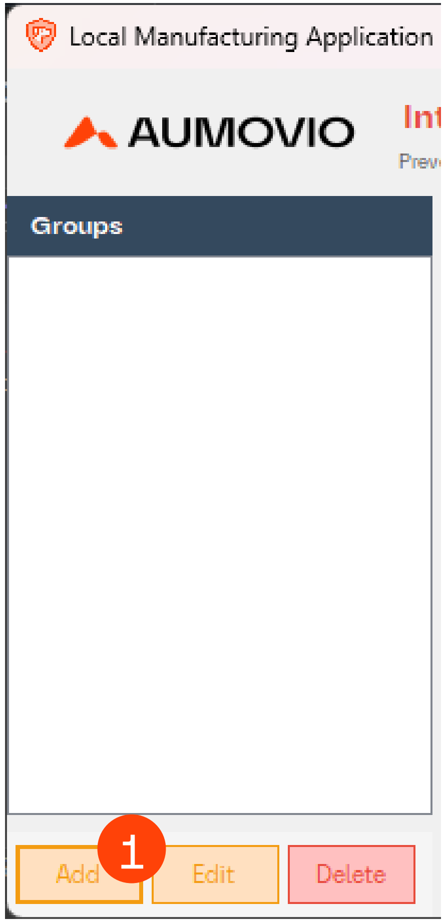
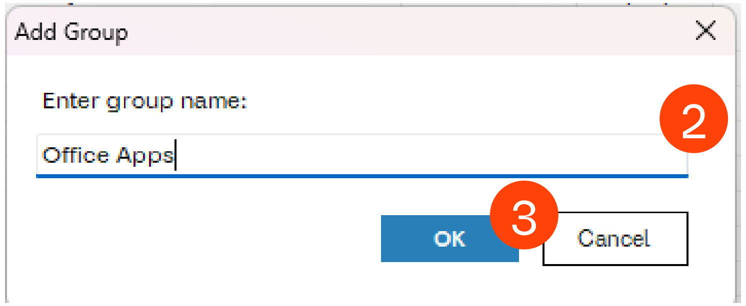
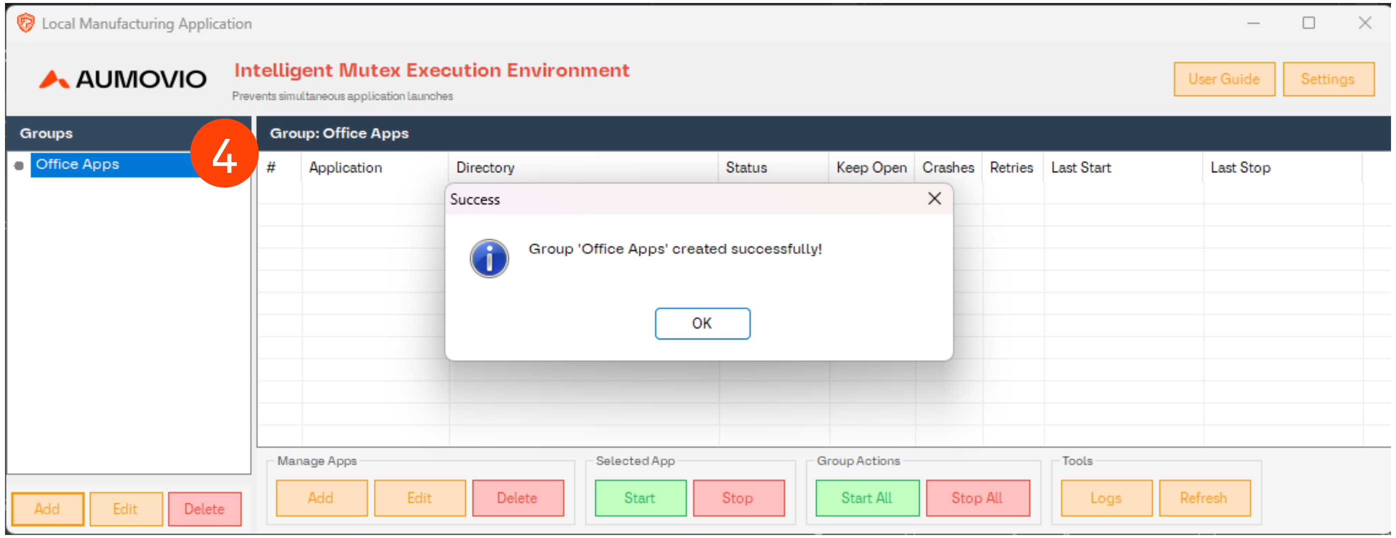
Managing Groups
Groups let you organize your applications logically







Create a Group

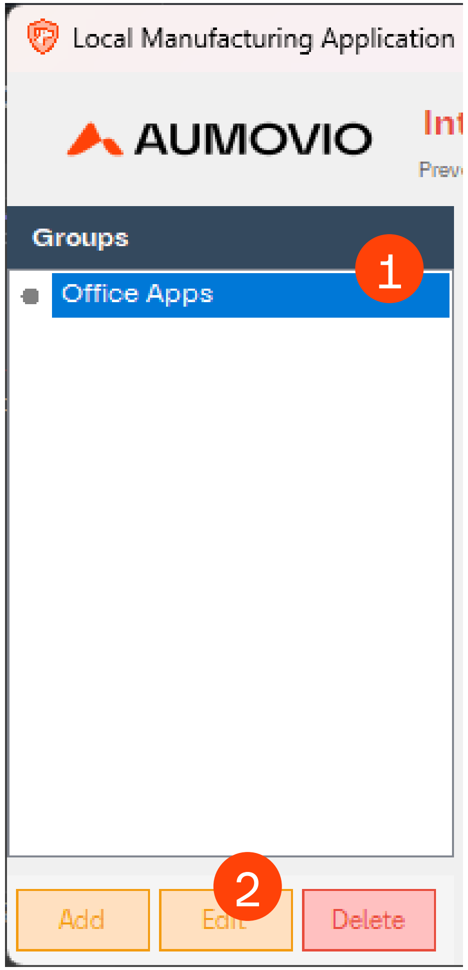
Step 1: Click **Add Group**

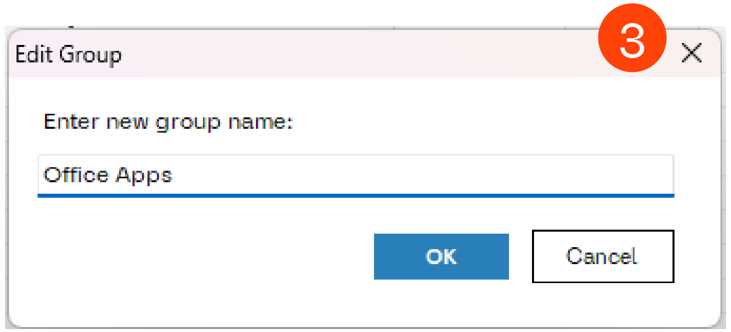
Step 2: Enter a name (e.g “CCU Top”, “ADAS SRR TOP”)

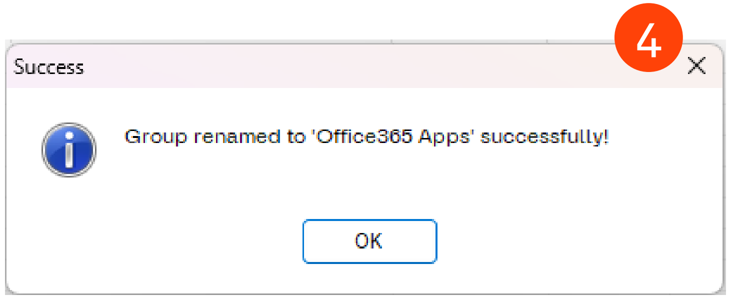
Step 3: Click OK

Step 4: The new group is created and automatically selected

Managing Groups
Groups let you organize your applications logically

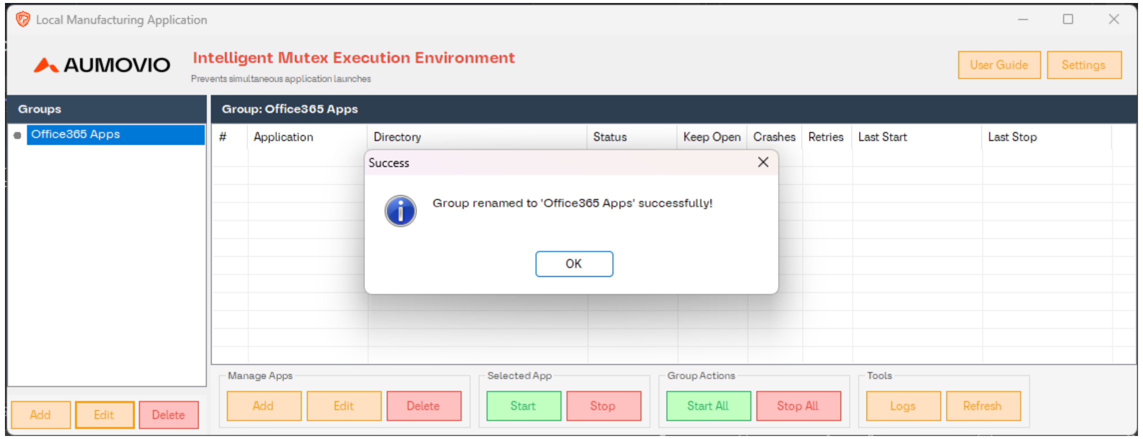




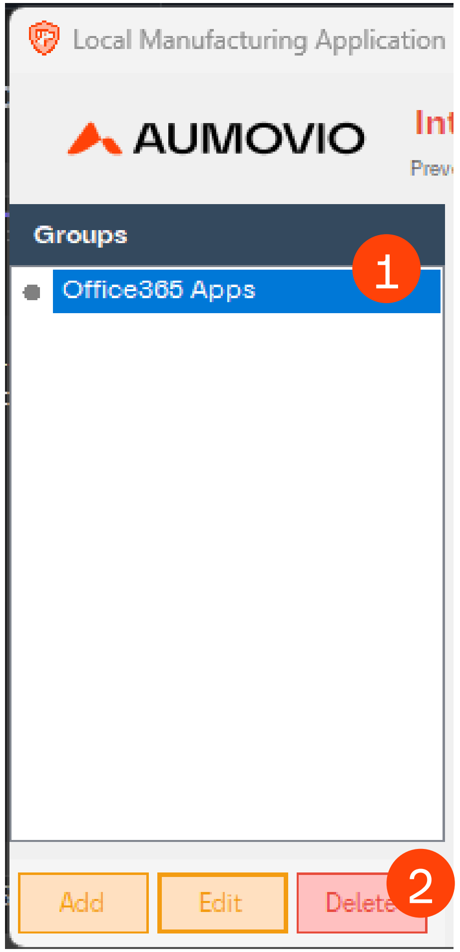


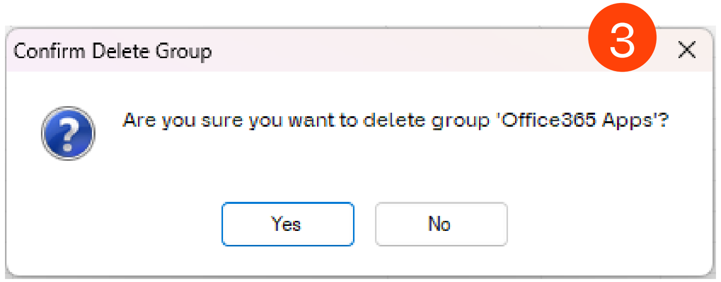
Rename a Group

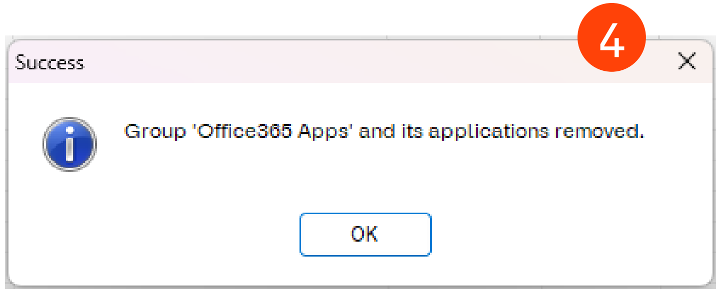
Step 1: Select the group in the left panel
Step 2: Click Edit
Step 3: Enter the new name
Step 4: Click Ok



Managing Groups
Groups let you organize your applications logically



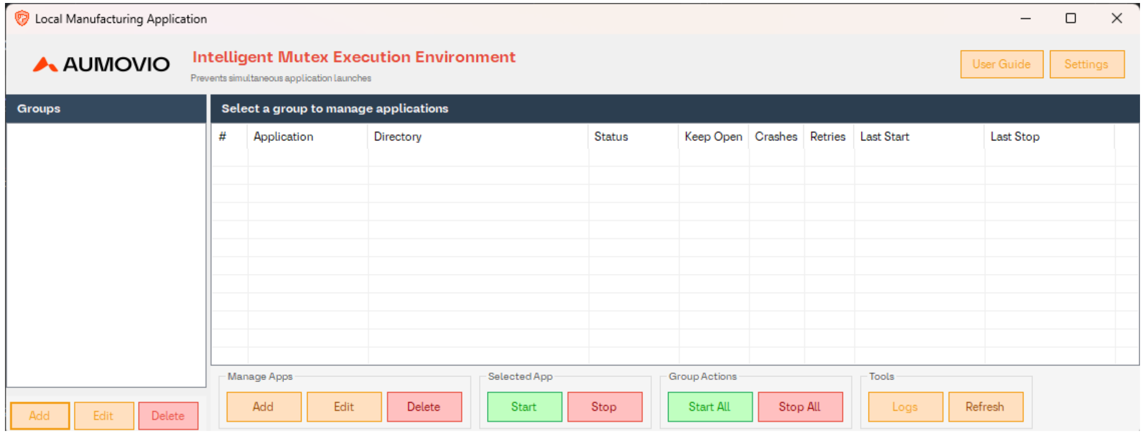




Note: Deleting a group does **not** delete the actual application files from disk.

Delete a Group
Step 1: Select the group in the left panel
Step 2: Click Delete
Step 3: Confirm the Deletion
Step 4: Click Ok

Important: All running applications in the group will be stopped, and all applications will be removed from management



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Managing Application for IMEE

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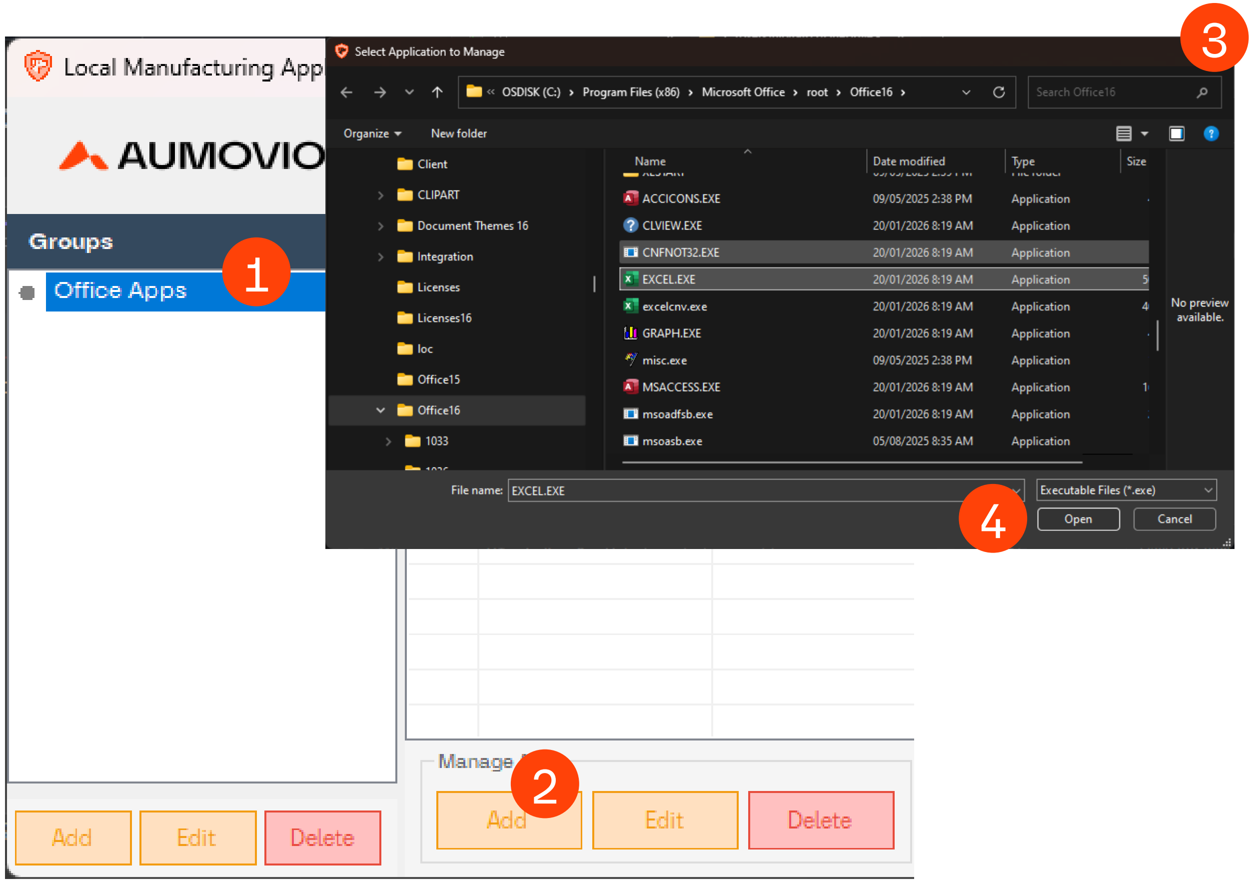
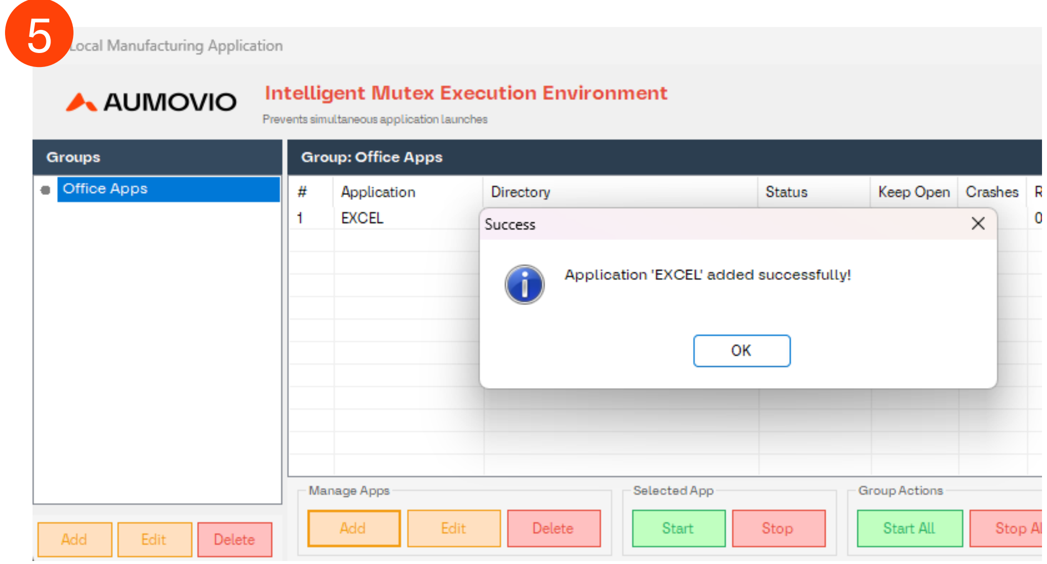
Managing Applications

Let's you organize your applications logically

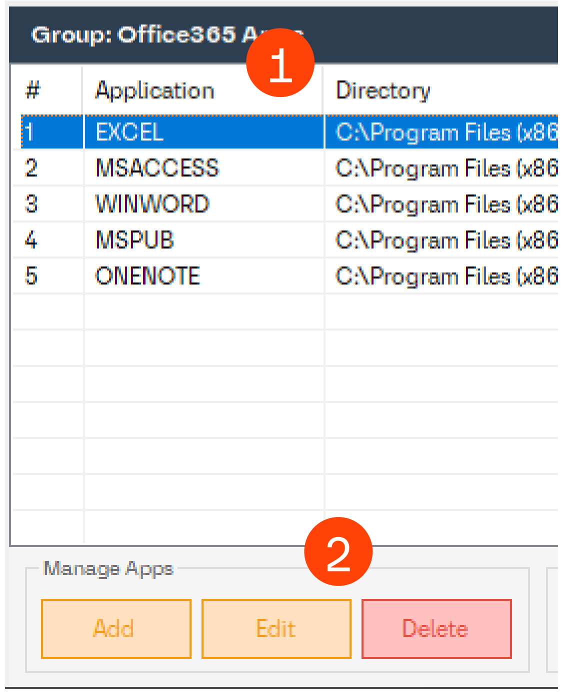


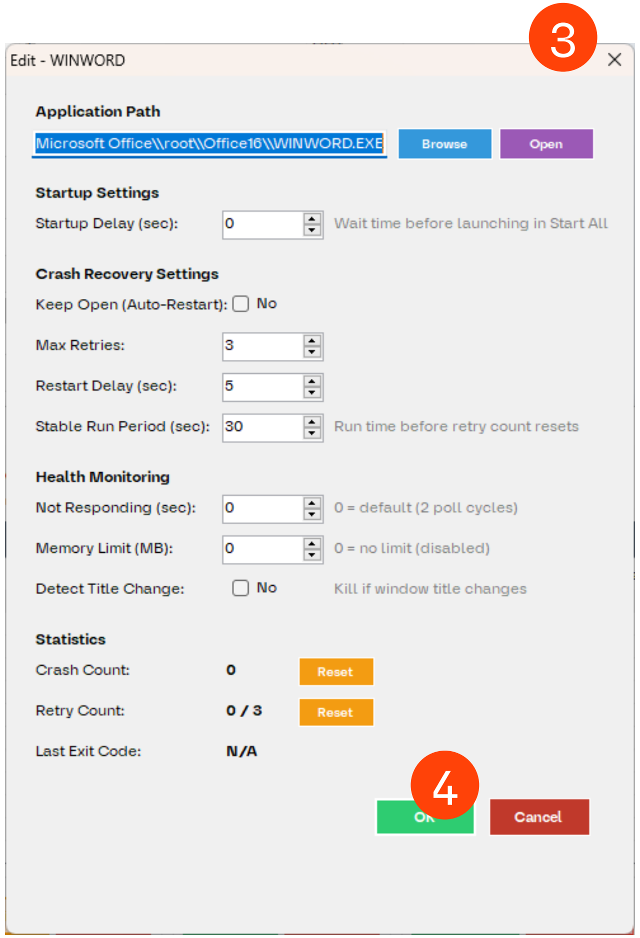
Adding an Application

Step 1: Select a group from the left panel
Step 2: Click Add
Step 3: Browse the executable (.exe) file
Step 4: Click open
Step 5: The application appears in the list with status "Stopped"



Managing Applications
Let’s you organize your applications logically



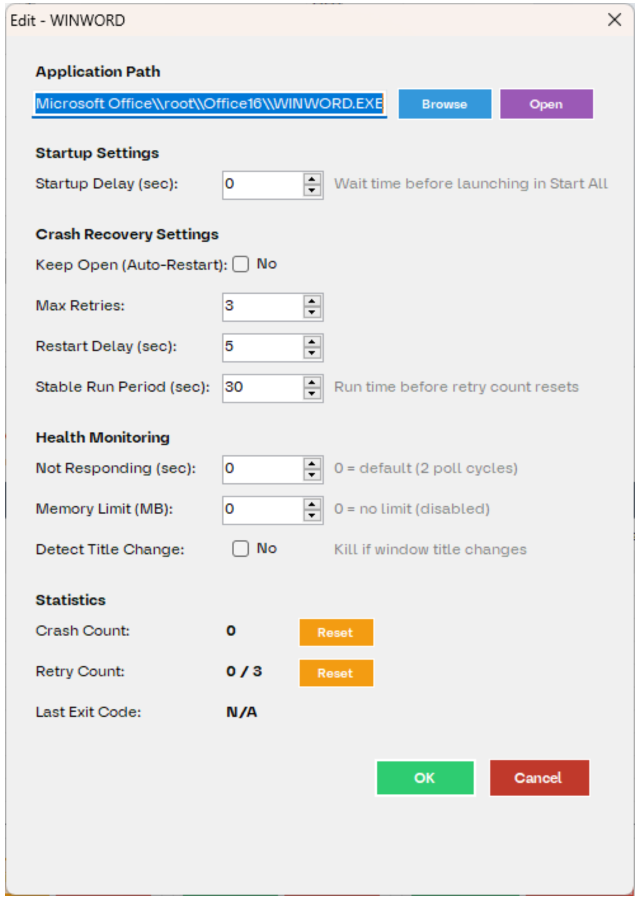


Editing an Application

Step 1: Select an application in the list
Step 2: Click Edit
Step 3: In the edit dialog, you can. See slide 16 to 20 on what each settings mean.
Step 4: Click OK to save changes

Step 4: Click OK to save changes

Managing Applications
Let’s you organize your applications logically



Application Path

- **Directory:** The full path to the executable. Use **Browse** to change it, or **Open** to reveal the file in File Explorer.

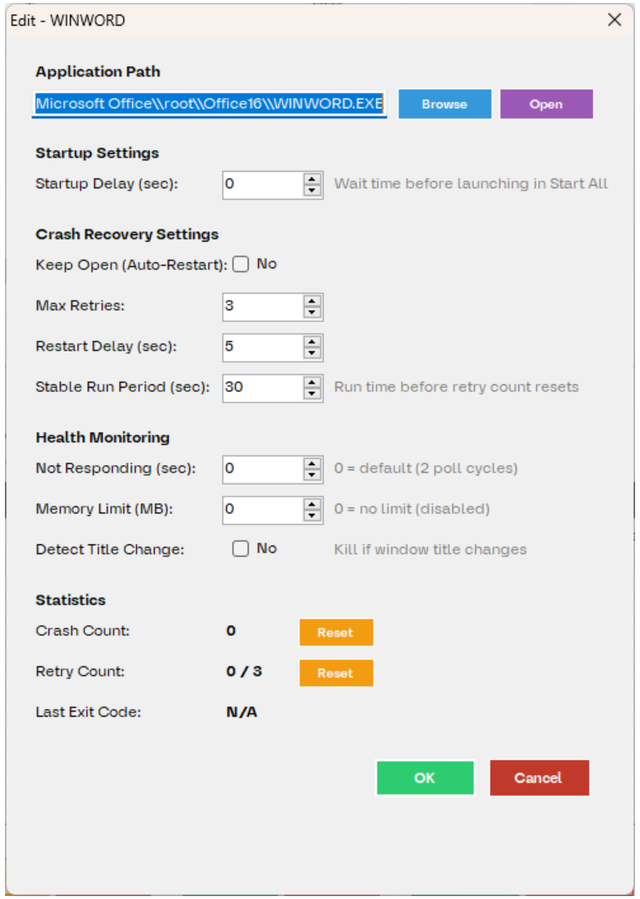
Startup Settings

Setting	Range	Default	Description
Startup Delay (sec)	0–300	0	Wait time before launching this app during sequential Start All. Apps with 0 launch immediately.

Crash Recovery Settings

Setting	Range	Default	Description
Keep Open (Auto-Restart)	Yes/No	No	When enabled, the watchdog automatically restarts the app after a crash.

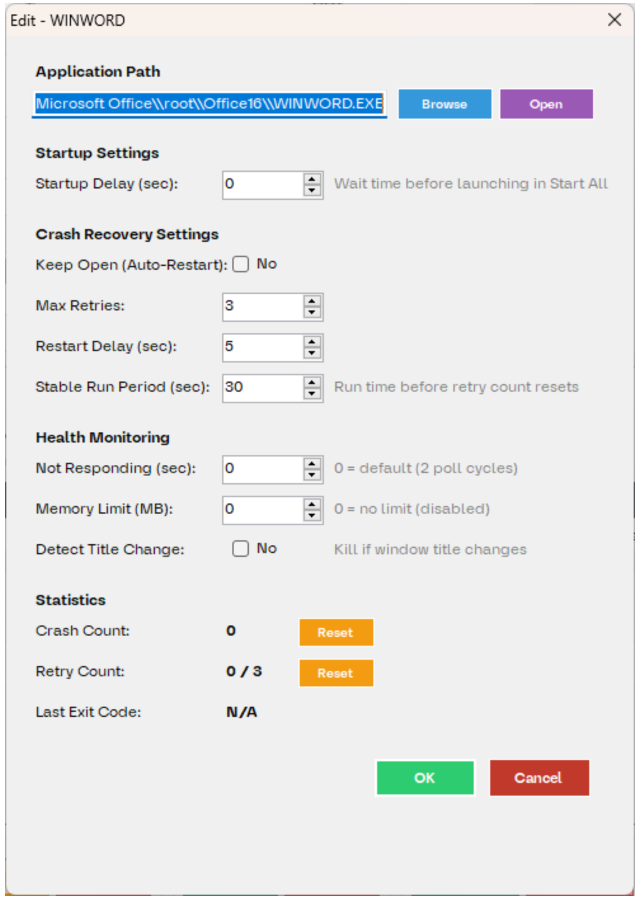
Managing Applications
Let’s you organize your applications logically



Crash Recovery Settings

Setting	Range	Default	Description
Max Retries	1–100	3	Maximum number of consecutive restart attempts before giving up.
Restart Delay (sec)	1–300	5	Seconds to wait before attempting a restart after a crash.
Stable Run Period (sec)	5–600	30	How long the app must run continuously after restart before the retry count resets to 0.

Managing Applications
Let’s you organize your applications logically

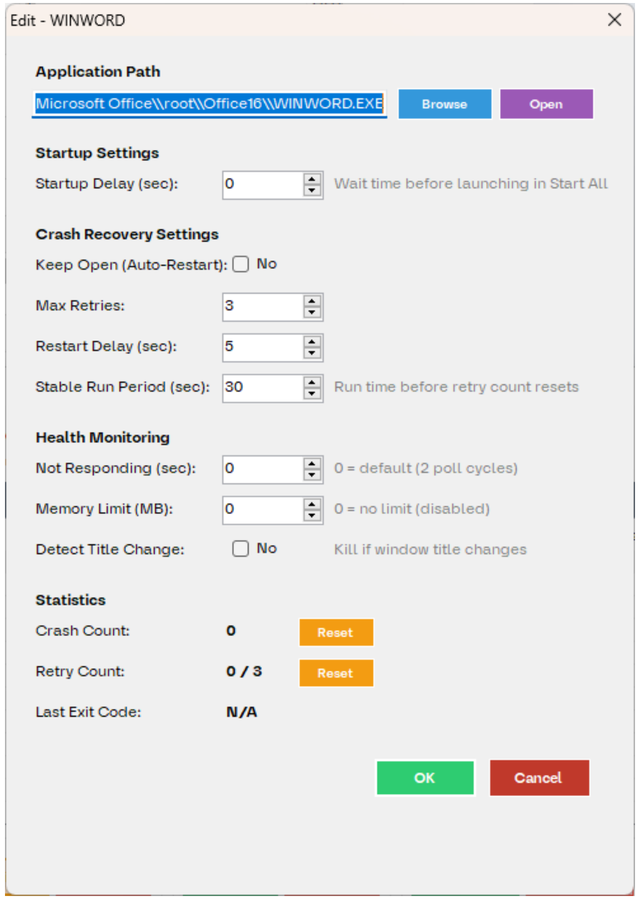


Health Monitoring

Setting	Range	Default	Description
Not Responding (sec)	0–600	0	How long a process can remain unresponsive before being force-killed. 0 = default behavior (2 consecutive poll cycles). Set a higher value for apps that legitimately freeze briefly during heavy operations.
Memory Limit (MB)	0–65536	0	Maximum working set memory allowed. If the process exceeds this limit, it is force-killed for auto-restart. 0 = no limit (disabled). Useful for detecting memory leaks in long-running apps.

Managing Applications

Let’s you organize your applications logically

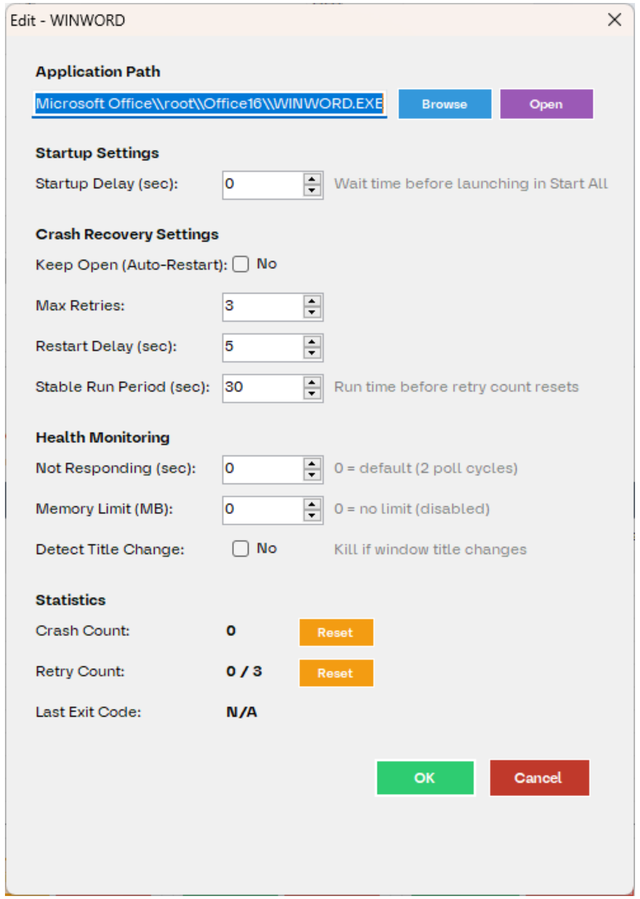


Health Monitoring

Setting	Range	Default	Description
Detect Title Change	Yes/No	No	When enabled, the watchdog records the app's initial window title and treats a persistent title change as a suspected error dialog (e.g., a WinForms unhandled exception dialog whose title is just the app name). Requires 2 consecutive detections to confirm. Leave disabled for apps that legitimately change their window title (e.g., showing a document name).

Managing Applications

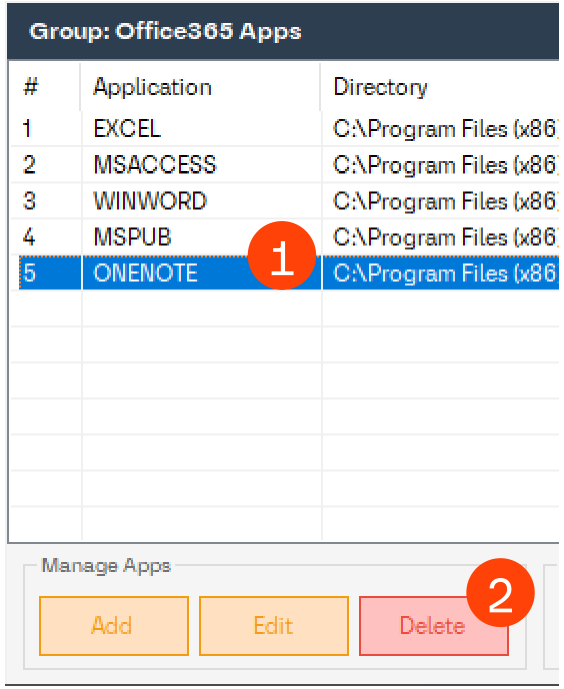
Let’s you organize your applications logically

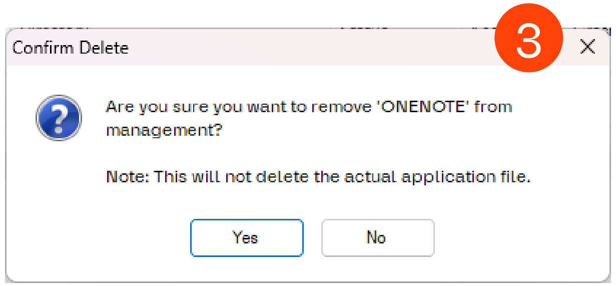


Statistics

Field	Description
Crash Count	Total number of detected crashes (cumulative). Click **Reset** to clear
Retry Count	Current consecutive restart attempt count vs. max retries (e.g., "1 / 3"). Click **Reset** to clear and remove "Failed" state.
Last Exit Code	The exit code from the last process termination. N/A if no exit has been recorded. A non-zero exit code (shown in red with "abnormal") typically indicates the process crashed or terminated with an error.

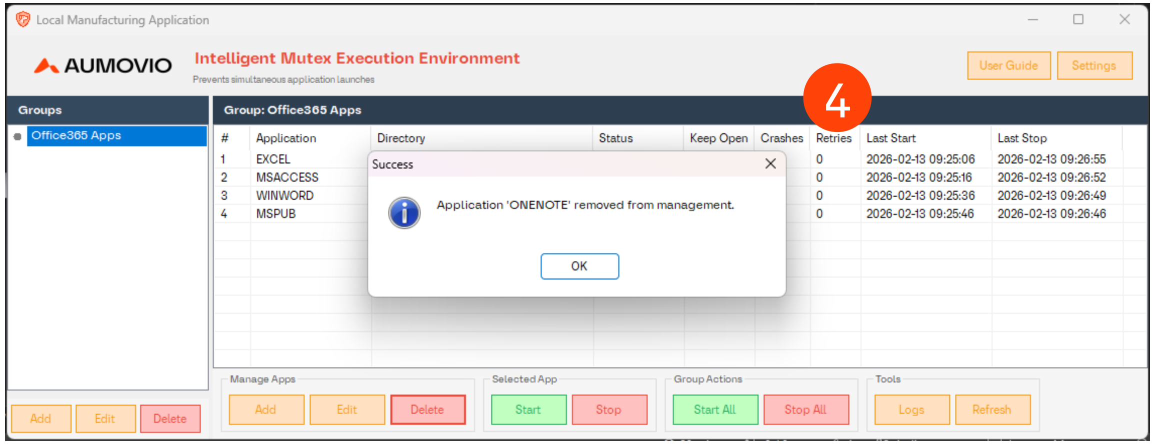
Managing Applications
Let’s you organize your applications logically





Deleting an Application
Step 1: Select an application in the list
Step 2: Click Delete
Step 3: Confirm deletion
Step 4: The application is removed from the management

Note: The actual executable file is not deleted from disk



Starting and Stopping Application

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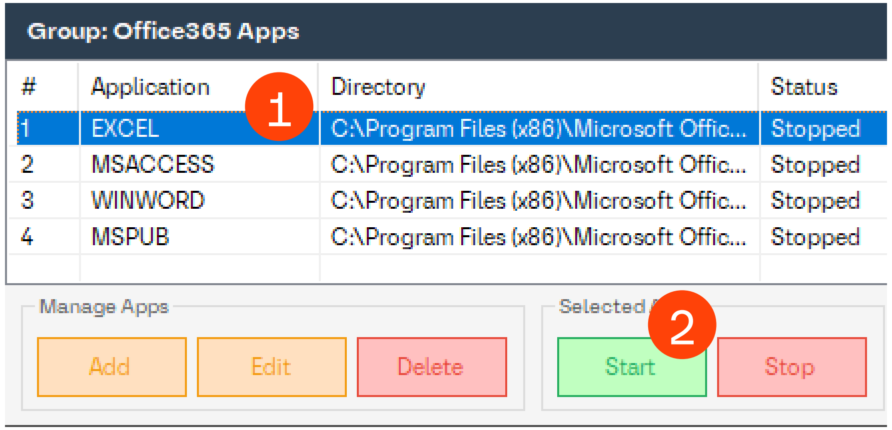
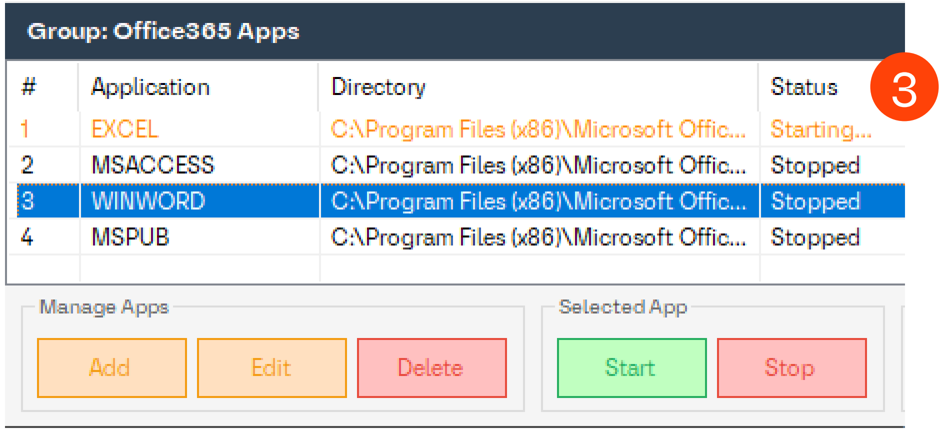
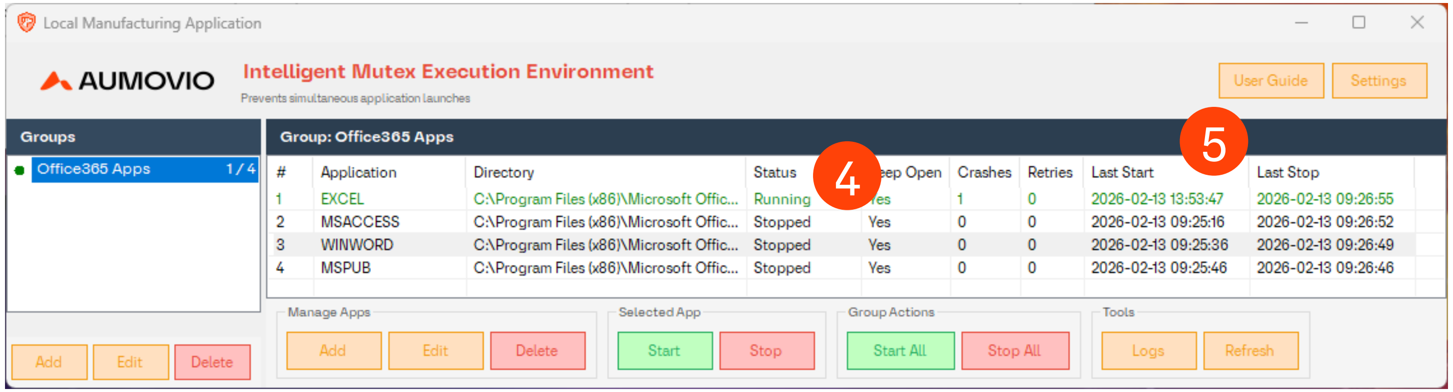
Starting and Stopping Application
Starting a Single Application

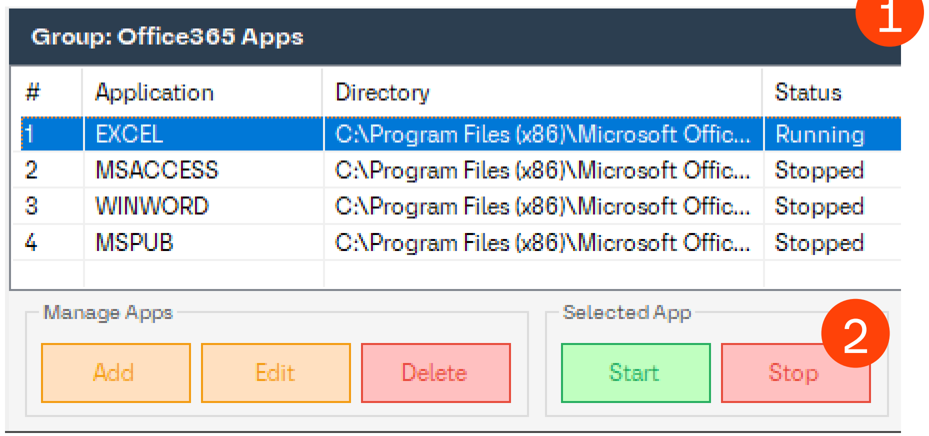


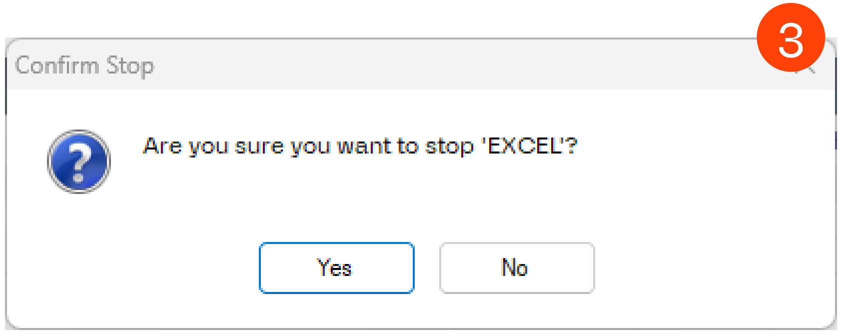


Start an Application
Step 1: Select an application from the list
Step 2: Click Start
Step 3: The status changes to **Starting...** (orange)
Step 4: Once the application window appears, the status changes to Running (green)
Step 5: The “Last Start” timestamp is recorded



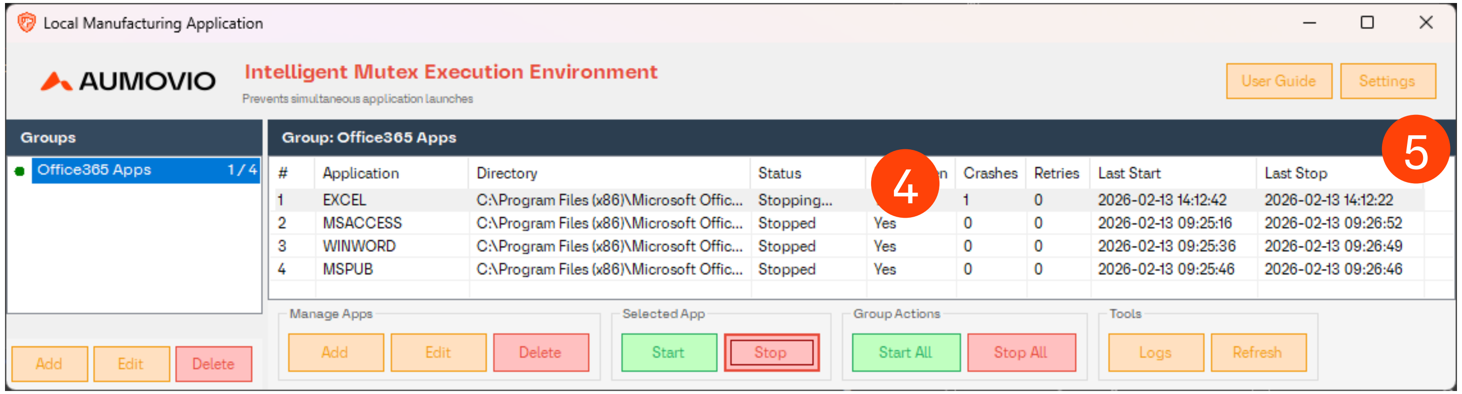
Starting and Stopping Application
Stopping a Single Application



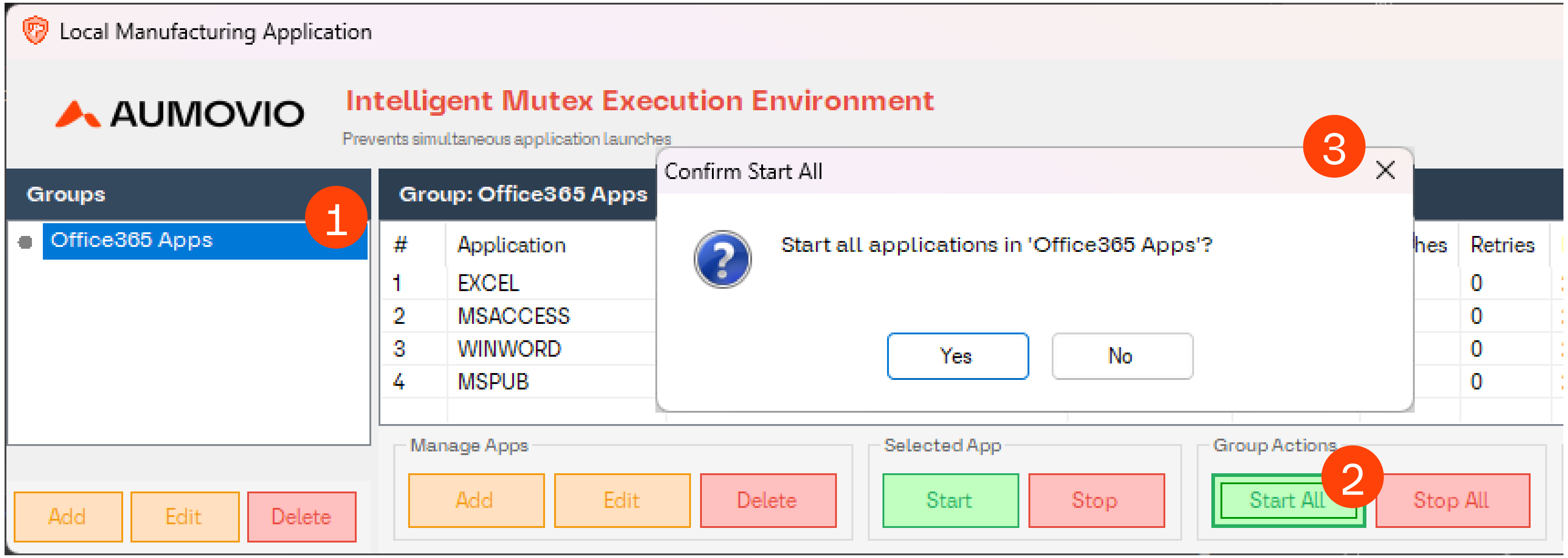


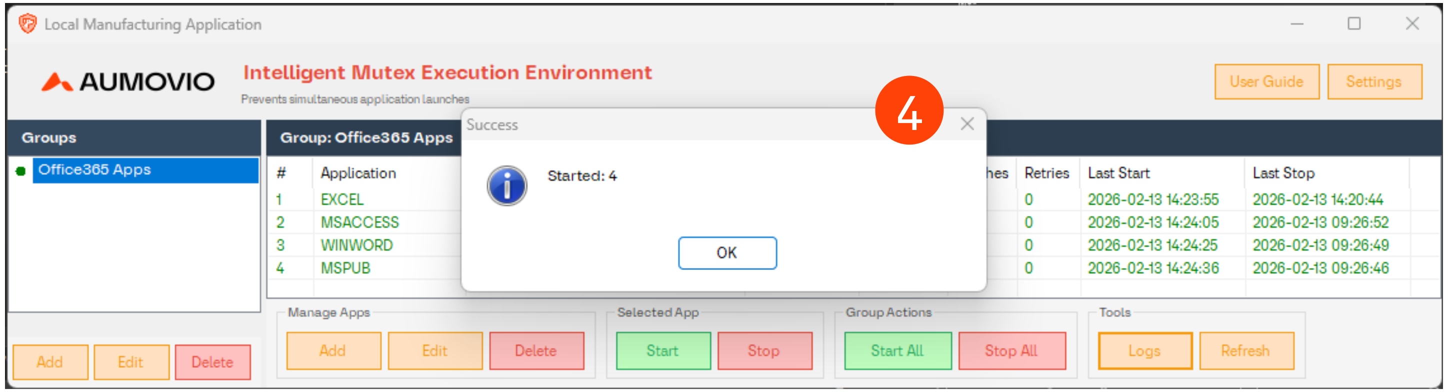
Stop an Application
Step 1: Select a running application
Step 2: Click Stop
Step 3: Confirm the action
Step 4: The status changes to **Stopping...** (orange)
Step 5: Once the process exits, the status changes to **Stopped** (black)
Step 6: The "Last Stop" timestamp is recorded

IMEE attempts a graceful shutdown first. If the application doesn't close within 3 seconds, it is force-killed.



Starting and Stopping Application
Starting All Application

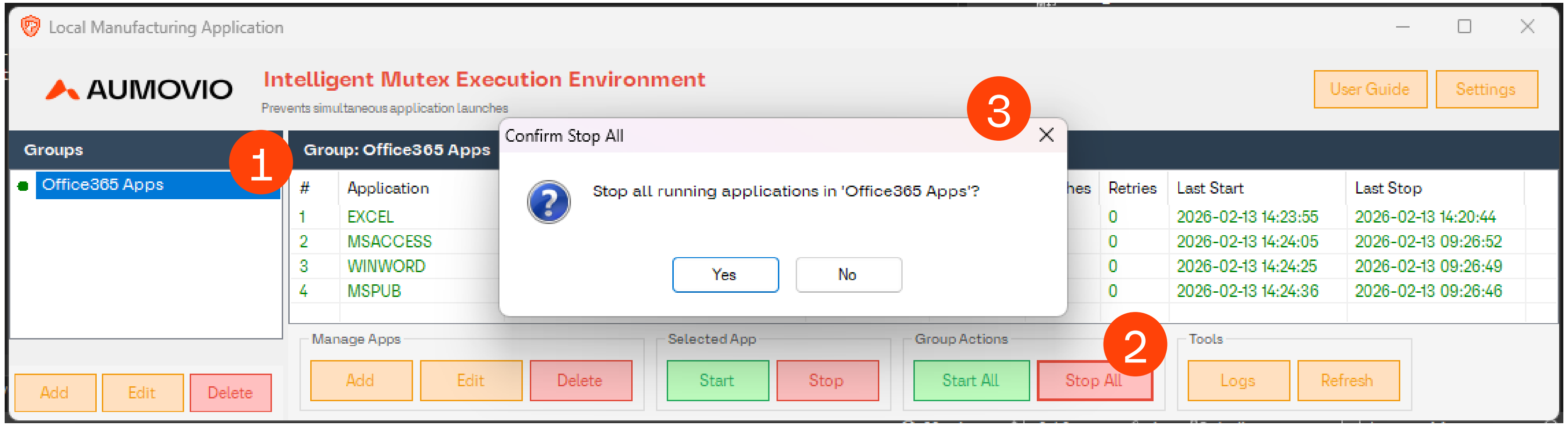


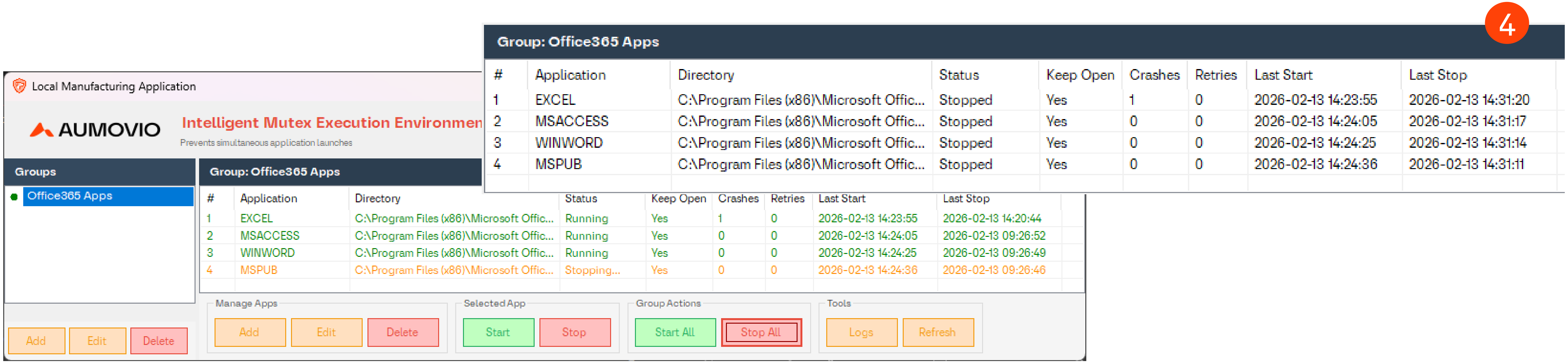


Starting All Application
Step 1: Select a Group
Step 2: Click Start All
Step 3: Confirm the action
Step 4: All stopped applications in the group are launched

With Startup Delays: If any applications have a Startup Delay configured (via Edit), they launch one at a time. Each app shows a "Waiting (Ns)" countdown before it launches. This prevents overloading the system when starting many apps at once.

Starting and Stopping Application
Stopping All Application





#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	1	0	2026-02-13 14:23:55	2026-02-13 14:31:20
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 14:24:05	2026-02-13 14:31:17
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 14:24:25	2026-02-13 14:31:14
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 14:24:36	2026-02-13 14:31:11

Stopping All Application

Step 1: Select a Group
Step 2: Click Stop All
Step 3: Confirm the action
Step 4: All running applications in the group are stopped

Keep Open (Auto-Restart)

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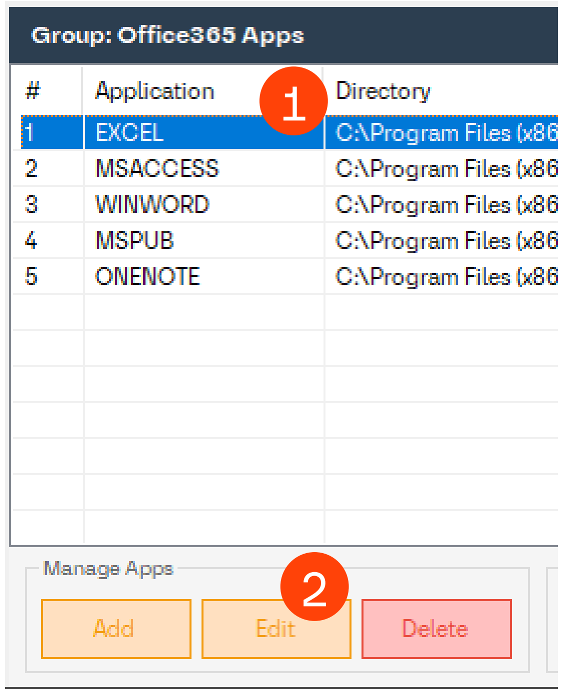
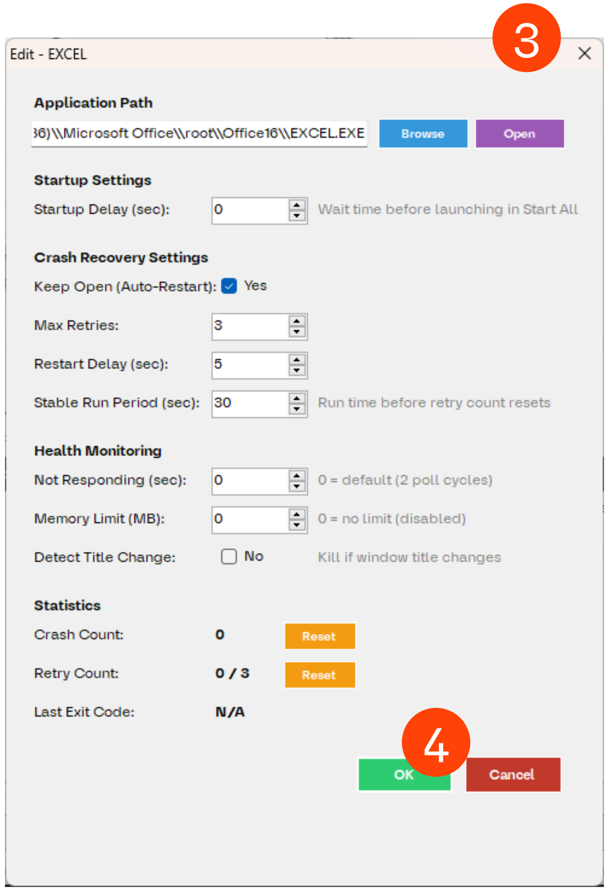
Keep Open (Auto-Restart)

Keep Open automatically restarts an application when it crashes or is closed unexpectedly.





Enabling Keep Open

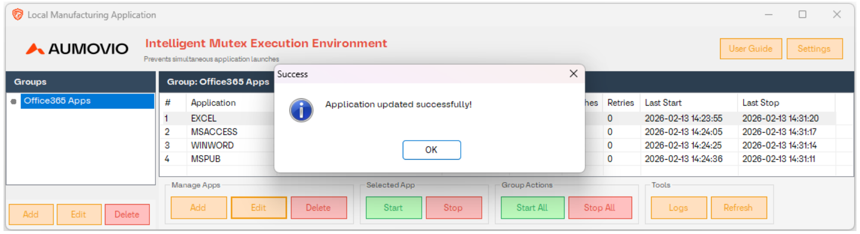
Step 1: Select an application in the list

Step 2: Click Edit

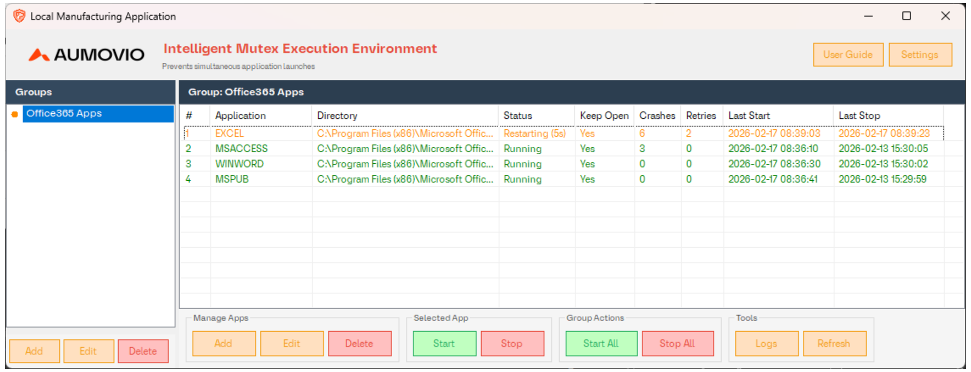
Step 3: Configure the settings:

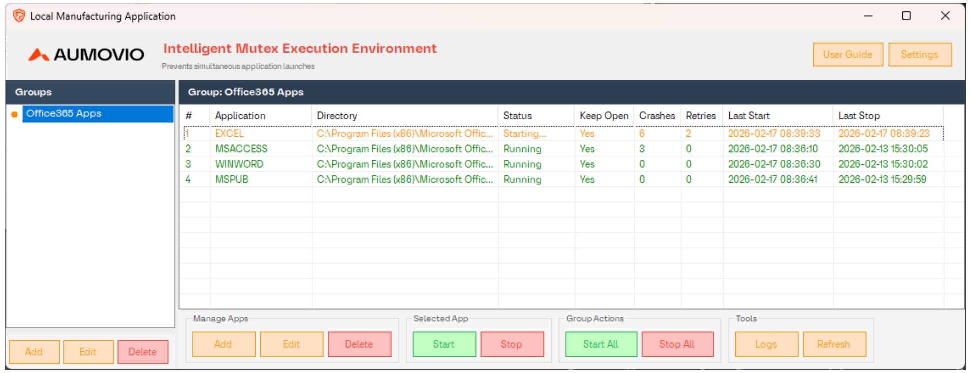
- Check Keep Open
- Restart Delay: How long to wait before restarting (default: 5 seconds)
- Max Retries: How many times to attempt restarting (default: 3)
- Stable Run Period: How long the app must run before retry count resets

Step 4: Click OK



Keep Open (Auto-Restart)
How it works

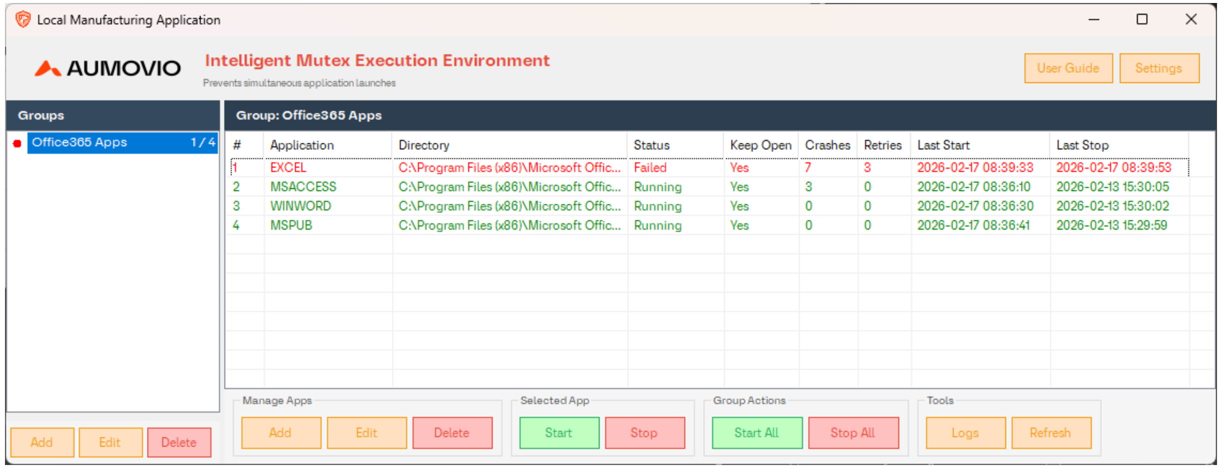




When a Keep Open application stops unexpectedly:

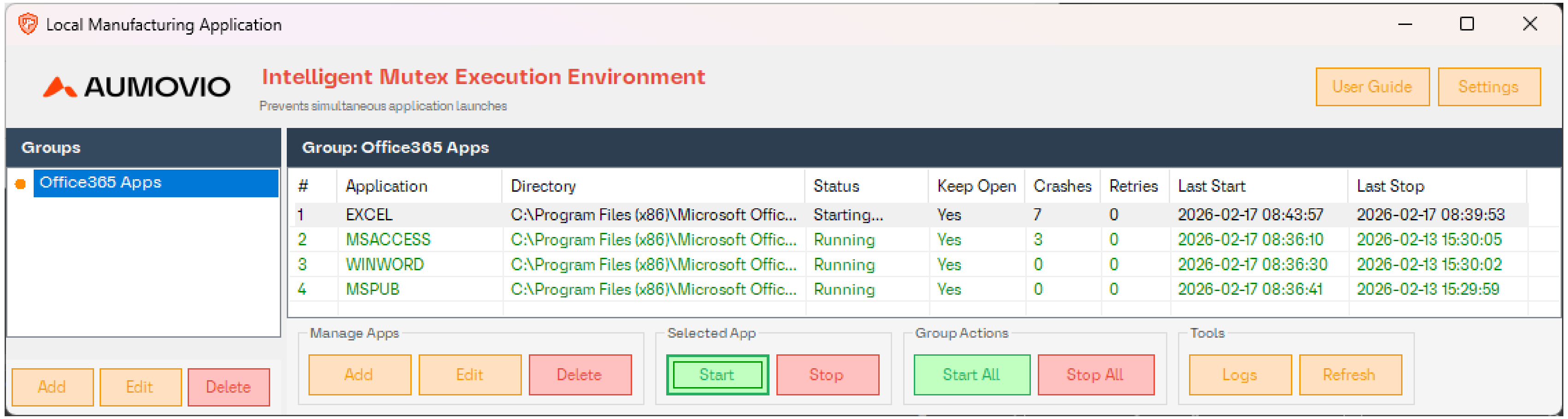
1. The crash is detected by the watchdog (within one poll interval, default: 10 seconds)
2. The crash count increments
3. The retry count increments
4. A countdown begins: **Restarting (5s)** ? **Restarting (4s)** ? … ? **Starting…**
5. The application relaunches automatically
6. If the relaunch succeeds and the app runs stably for the **Stable Run Period** (default: 30 seconds), the retry count resets to 0

Keep Open (Auto-Restart)
When Max Retries Are Exhausted



If the application crashes more times than the max retry limit (without running stably between crashes):

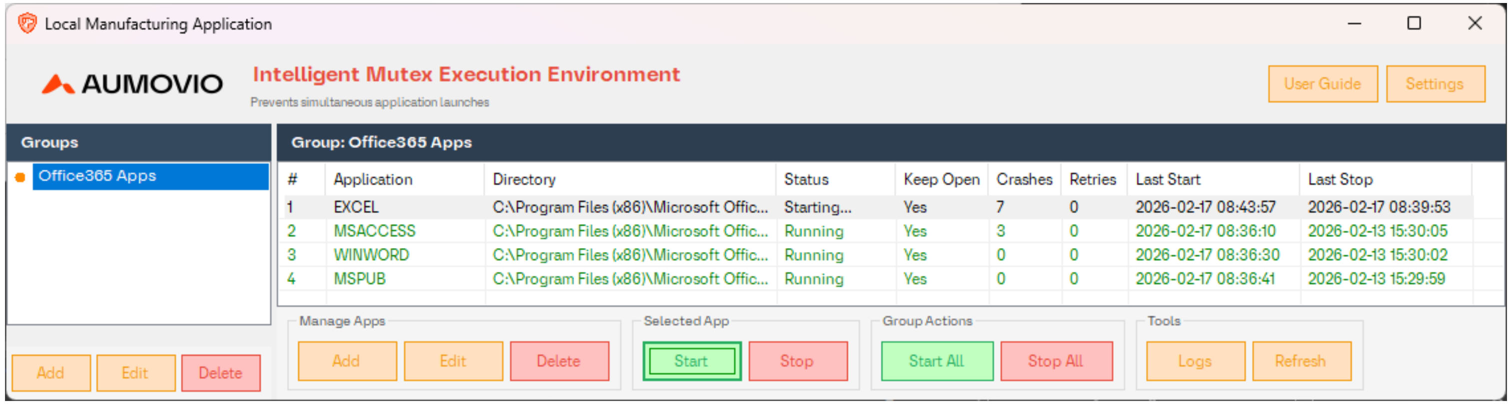
1. The status changes to **Failed** (red)
2. No further auto-restart attempts are made
3. You can fix the issue and either:
   - Click **Start** to manually restart (resets the retry count)
   - Click **Edit** and reset the retry count, then start



Keep Open (Auto-Restart)
Stable Run Period

The retry count only resets to 0 after the application has been running continuously for the configured **Stable Run Period** (default: 30 seconds). This prevents apps that crash shortly after starting from resetting retries indefinitely and never reaching "Failed" state.

You can configure this per-application via **Edit ? Stable Run Period (seconds).** [see slide 13]

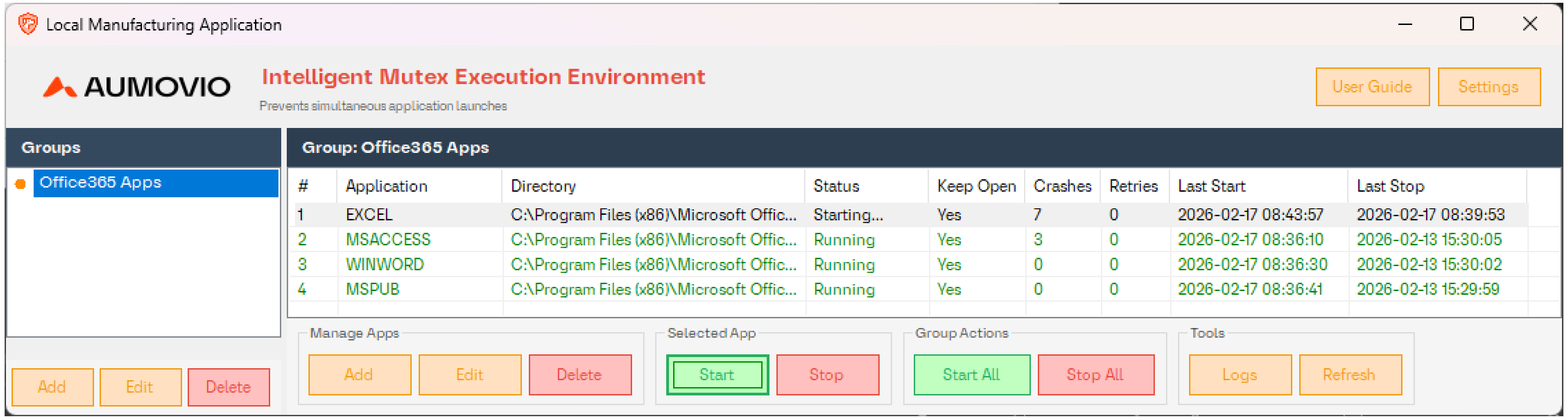


Keep Open (Auto-Restart)
Unresponsive Application Detection

If a Keep Open application becomes unresponsive (e.g., due to an unhandled exception dialog that blocks the main window), Instance Manager will detect it and force-kill the process so auto-restart can take over. The detection requires two consecutive poll cycles to confirm the app is truly unresponsive, avoiding false positives from brief UI freezes.

Important Notes

– Manually stopping an app via the **Stop** button does **not** trigger a restart – even with Keep Open enabled
– The retry count resets to 0 whenever the app successfully starts (manual or auto)
– The crash count is cumulative and never resets automatically (reset it via Edit)



AUMOVIO

Health Monitoring

Complete Application User Guide

IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Health Monitoring
Not Responding Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Not Responding Detection
Detects when an application's UI thread is frozen (e.g., infinite loop, UI deadlock, blocked message pump).

How it works:
1. Each poll tick checks Process.Responding for all windowed processes
2. If an app is not responding, tracking begins
3. After the configured timeout (or 2 consecutive poll cycles if timeout is 0), the process is force-killed
4. Auto-restart takes over

Configure via: Edit → **Not Responding (sec)**
Tip: Set a non-zero value (e.g., 30 seconds) for apps that occasionally freeze briefly during heavy operations like loading large files.

Health Monitoring
Error Dialog Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Error Dialog Detection
Detects when an application shows a crash or error dialog that is technically "responding" (pumping messages) but has the app effectively stuck. This catches error dialogs that Process.Responding cannot detect because the dialog's message pump keeps the window alive.

Detected patterns include:
- "has stopped working", "has encountered a problem"
- "unhandled exception", "application error", "runtime error", "fatal error"
- ".NET Framework", "CLR error", "just-in-time debugging"
- "assertion failed", "access violation", "stack overflow"
- .NET exception type names
(e.g., NullReferenceException, DivideByZeroException, OutOfMemoryException, StackOverflowException, AccessViolationException, InvalidOperationException)
- Generic phrases like "an error occurred", "error in application"

Health Monitoring
Error Dialog Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

How it works:
1. Each poll tick checks the main window title against known error dialog patterns
2. If a match is found, it waits one more poll cycle to confirm (avoids false positives)
3. On second consecutive detection, the process is force-killed
4. Auto-restart takes over

This detection is **always active** for Keep Open apps – no configuration needed.

Health Monitoring
Window Title Change Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Window Title Change Detection
Detects when an application's window title changes unexpectedly, which can indicate a WinForms ThreadExceptionDialog or similar error dialog whose title doesn't contain any recognizable error keywords.

How it works:
1. When an app first appears as running, its window title is recorded as the baseline
2. Each poll tick compares the current title to the baseline
3. If the title changes and stays changed for 2 consecutive poll cycles, the process is force-killed
4. Auto-restart takes over

Configure via: Edit → **Detect Title Change** (opt-in, disabled by default)

Warning: Only enable this for apps with a stable, unchanging window title. Apps that show document names, status text, or progress in their title bar will trigger false positives.

Health Monitoring
Memory Limit Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Memory Limit Detection
Detects when an application's working set memory exceeds a configured threshold. Useful for catching memory leaks in long-running 24/7 applications.

How it works:
1. Each poll tick reads Process.WorkingSet64 (a lightweight kernel query with no overhead)
2. If the working set exceeds the configured limit, the process is force-killed immediately (no confirmation wait)
3. Auto-restart takes over

Configure via: Edit → **Memory Limit (MB)** (set to 0 to disable)

Tip: Monitor your app's normal memory usage first, then set the limit to 2–3× the expected peak. For example, if an app normally uses 200 MB, set the limit to 500 MB.

Health Monitoring
CPU-Hung Process Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

CPU-Hung Process Detection
Detects when an application is consuming excessive CPU continuously, indicating an infinite loop or spin-wait deadlock.

How it works:
1. Each poll tick samples Process.TotalProcessorTime (a lightweight kernel query)
2. CPU utilization is computed by comparing samples across ticks: ΔCpuTime / (Δtime × ProcessorCount)
3. If CPU usage exceeds 95% for 3 consecutive poll cycles, the process is force-killed
4. Brief CPU spikes are tolerated – only sustained high CPU triggers the kill

This detection is **always active** for Keep Open apps – no configuration needed.

Note: This catches spin-wait deadlocks (high CPU) but not idle deadlocks (0% CPU with frozen UI). Idle deadlocks are caught by Not Responding Detection instead.

Health Monitoring
Background Zombie Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Background Zombie Detection
Detects when an application's window disappears but its process remains alive in the background (e.g., Excel closing its window but lingering as a background process).

How it works:
1. If the watchdog detects that an app was running (had a window) but now has no window, it checks for background processes with the same name
2. Any found background processes are force-killed
3. The app is then treated as crashed and the normal restart flow begins

This detection is **always active** – no configuration needed.

Health Monitoring
Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
UI Freeze	Infinite loop on UI thread	Not Responding detection	Yes
Unhandled Exception Dialog	NullReferenceException, DivideByZeroException on UI thread	Error Dialog title matching + Title Change detection	Yes
Stack Overflow	Infinite recursion	Process terminates → crash detection	Yes
Out of Memory	Memory exhaustion	Memory Limit detection (proactive) + crash detection (reactive)	Yes
Access Violation	Writing to invalid memory	Process terminates → crash detection	Yes

Health Monitoring
Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
UI Deadlock	Two locks acquired in opposite order	Not Responding detection (UI blocked) + CPU-Hung detection (spin-wait)	Yes
Background Thread Exception	Unhandled exception on worker thread	Process terminates → crash detection + zombie detection	Yes
Environment.FailFast	Immediate process termination	Process terminates → crash detection	Yes
Process Crash	Any unexpected process exit	Window disappears → crash detection	Yes
Memory Leak	Gradual memory growth	Memory Limit detection	Yes

Health Monitoring
Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
CPU Spin Loop	Infinite while(true) with work	CPU-Hung detection (3 consecutive ticks >95%)	Yes
Zombie Process	Window closed but process lingers	Background Zombie detection	Yes

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User Experience

Complete Application User Guide

IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Understanding Status Indicators
Application Status

Status	Color	What it Means	What to Do
Stopped	Black	Application is not Running	Click Start to launch
Starting...	Orange	Application was just launched, waiting for its window to appear	Wait – will change to Running
Running	Green	Application is running normally	No action needed
Running (Other Group)	Dark Cyan	Application is running, but it was started from a different group	Stop it in the other group first if you want to start it here
Stopping...	Orange	Stop command was sent, waiting for the process to exit	Wait – will change to Stopped
Restarting (Ns)	Orange	Application crashed, auto-restart countdown in progress	Wait for countdown to finish
Starting...	Orange	Restart countdown finished, application is being relaunched	Wait – will change to Running

Understanding Status Indicators
Application Status

Status	Color	What it Means	What to Do
Waiting (Ns)	Orange	Sequential Start All – this app is next in line	Wait for countdown to finish
Failed	Red	Auto-restart gave up after max retries	Check the app, then Start manually or reset retries

Note: Keep Open applications that become unresponsive (e.g., stuck on an unhandled exception dialog) are automatically detected and force-killed after two consecutive poll cycles, allowing auto-restart to take over.

Understanding Status Indicators
Group Status

Each group in the left panel displays a colored circle indicator that summarizes the state of all applications in that group. When only some applications are in a given state, a count suffix (e.g., "2 / 5") appears next to the group name.

Color	What it Means	Count suffix
Gray	Group has no applications, or all applications are stopped	N/A
Green	All applications are running	N/A
Green	Some applications are running	N / T (running / total)
Orange	One or more applications are in a transitional state (Starting, Stopping, Restarting or Waiting)	N/A
Red	All applications have failed (max retries exhausted)	N/A
Red	Some applications have failed	N / T (failed / total)

Priority: If a group has apps in multiple states, the indicator shows the highest-priority state: Red (failed) > Orange (transitional) > Green (running) > Gray (stopped/empty).

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Settings

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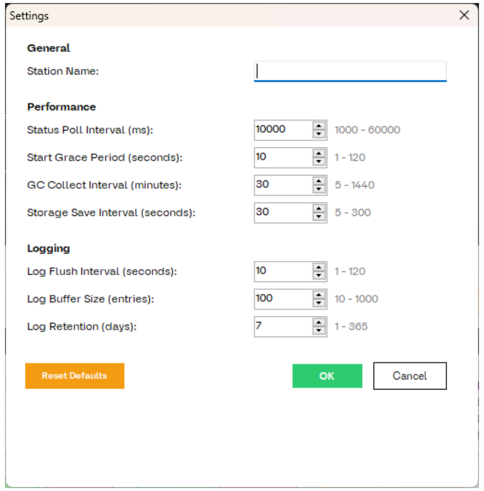
IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Settings

IMEE Settings and Performance tweaks



The Settings dialog lets you configure the station name and tune performance and logging parameters. Click the **Settings** button in the header to open it.

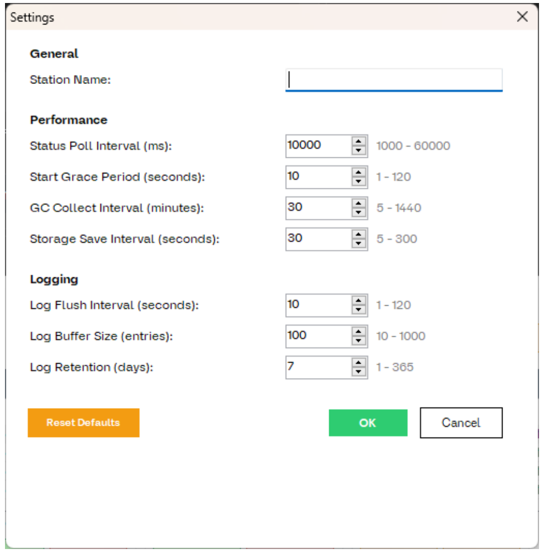
General

Setting	Description	Default
Station Name	Identifies the workstation in the title bar	(empty

Logging

Setting	Range	Default	Description
Log Flush Interval (seconds)	1 – 120	10	How often buffered log entries are written to disk
Log Buffer Size (entries)	10 – 1000	100	Maximum buffered entries before a flush is forced.
Log Retention (days)	1 – 365	7	Log files older than this are automatically deleted

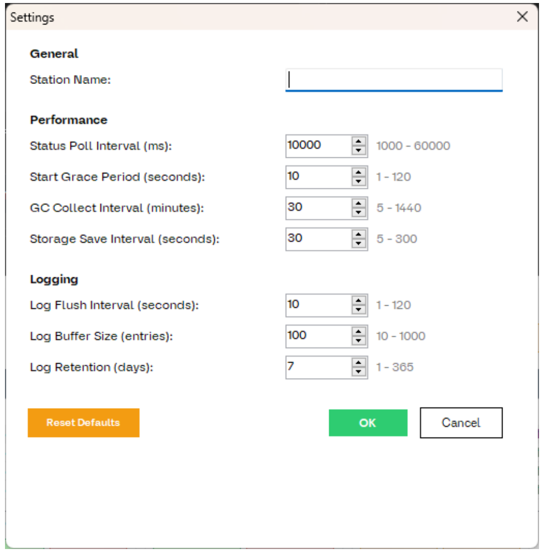
Settings
IMEE Settings and Performance tweaks



Performance

Setting	Range	Default	Description
Status Poll Interval (milliseconds)	1000 – 60000	10000	How often the watchdog checks process statuses. Lower = more responsive but higher CPU usage.
Start Grace Period (seconds)	1 – 120	10	Time after launching an app before the watchdog starts monitoring it. Allows the window to appear.
GC Collect Interval (minutes)	5 – 1440	30	How often a periodic garbage collection runs to prevent long-term memory growth during 24/7 operation.
Storage Save Interval	5 – 300	30	Minimum time between throttled disk writes for timer-tick updates. User-initiated saves (Add, Edit, Delete) always write immediately.

Settings
IMEE Settings and Performance tweaks



Reset Defaults
Click **Reset Defaults** in the Settings dialog to restore all performance and logging settings to their defaults. The station name is not affected by reset.

All settings are saved in `settings.json` and persist across restarts. Changes take effect immediately – no restart required.

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Logs

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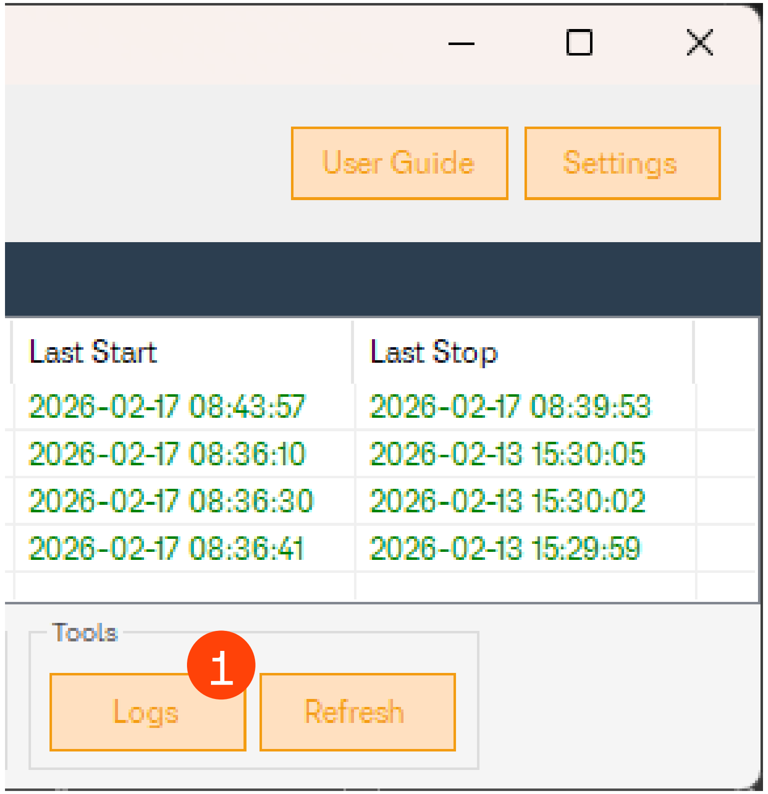
IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Logs
Instance Manager logs all operations to daily log files.

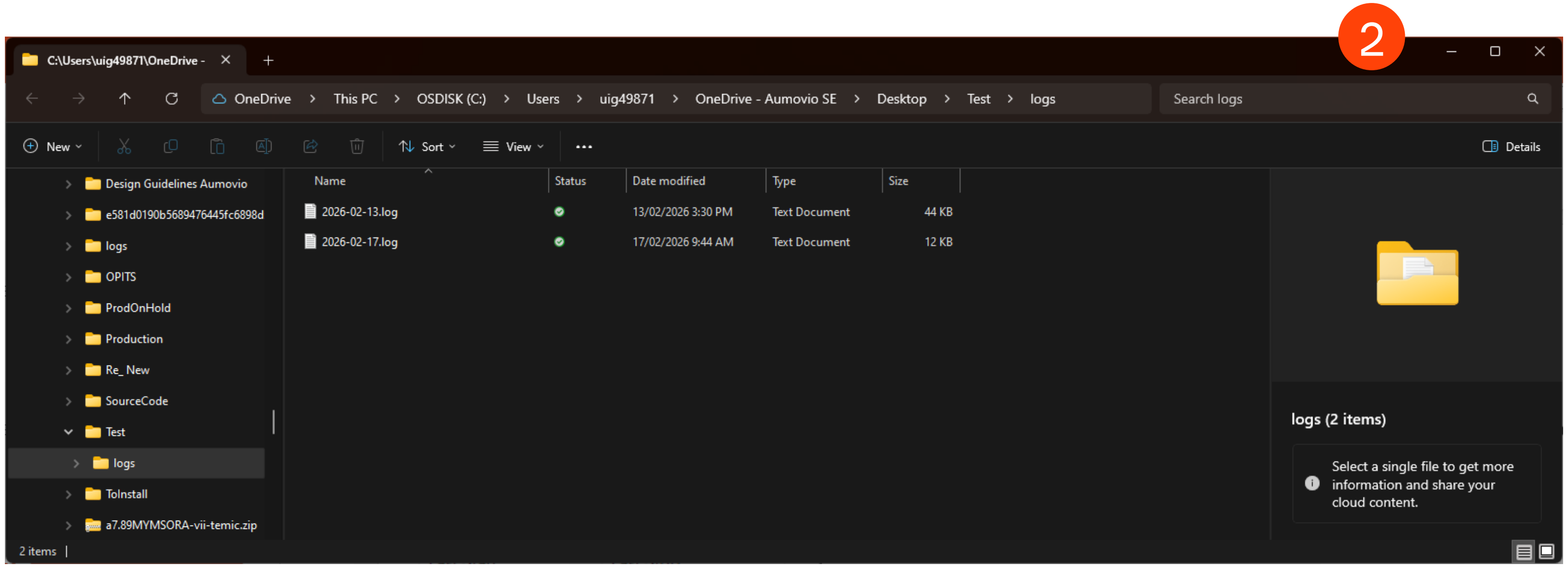


Viewing Logs

Step 1: Click the **Logs** button to open the logs folder in File Explorer

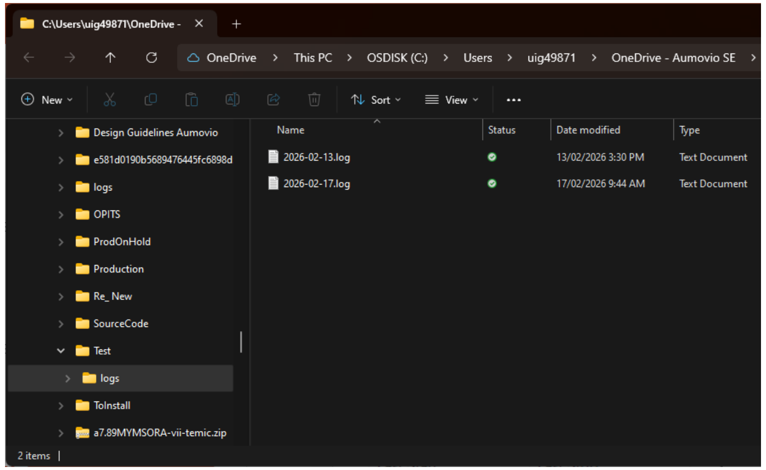
Step 2: You can now browse the logs

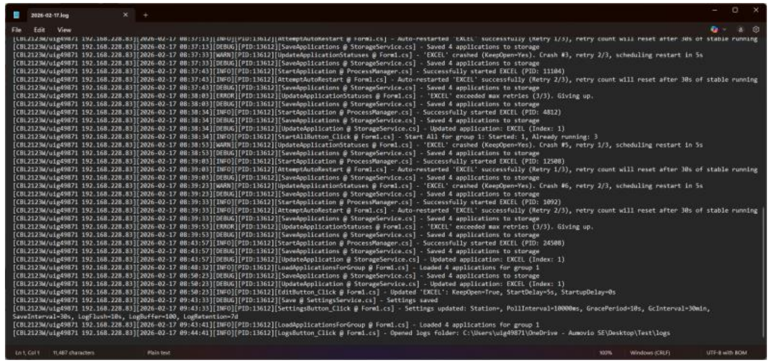
Log files are named `YYYY-MM-DD.log`



Logs

Instance Manager logs all operations to daily log files.





What Gets Logged

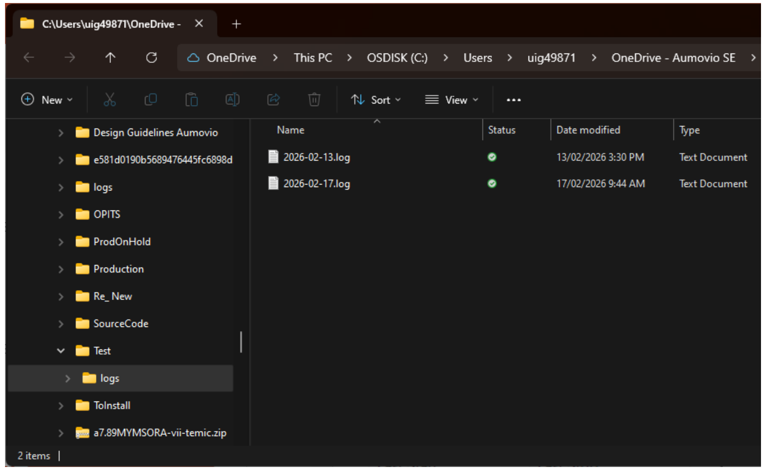
- Application starts and stops (including PIDs)
- Unauthorized launch detections
- Crash detections and auto-restart attempts
- Health monitoring events (not responding, error dialogs, memory limit breaches, CPU-hung detection, title changes)
- Group and application add/edit/delete operations
- Errors and warnings
- Startup terminations

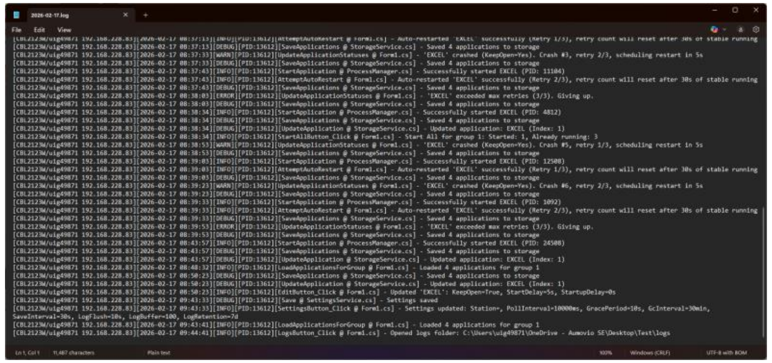
Log Format

[HOSTNAME/USER IP][2025-06-15 14:30:00][INFO][PID:12345][StartApplication @ ProcessManager.cs] - Successfully started notepad (PID: 9876)

Logs

Instance Manager logs all operations to daily log files.





Log Retention

Logs older than the configured retention period (default: 7 days) are automatically deleted. You can change this in **Settings ? Log Retention (days)**.

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Security

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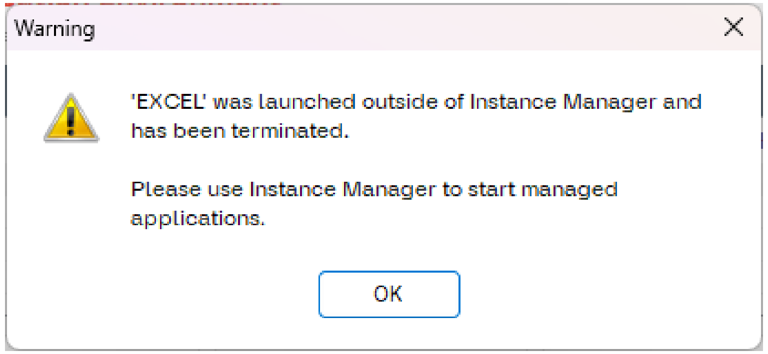
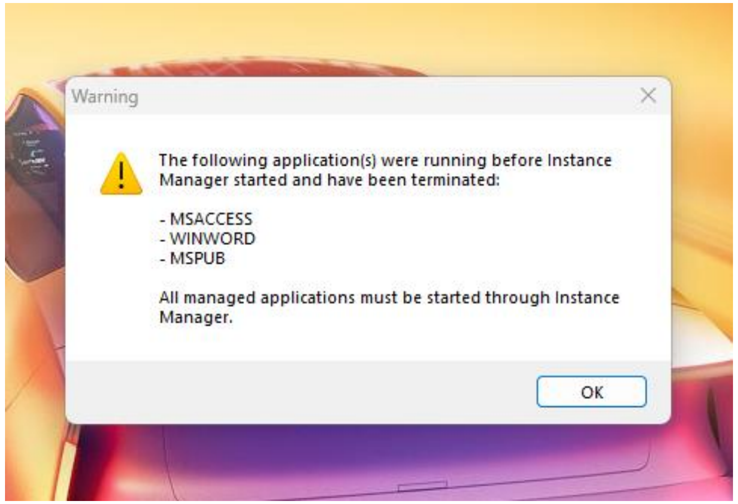
IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Security
Managed applications can **only** be launched through its interface.

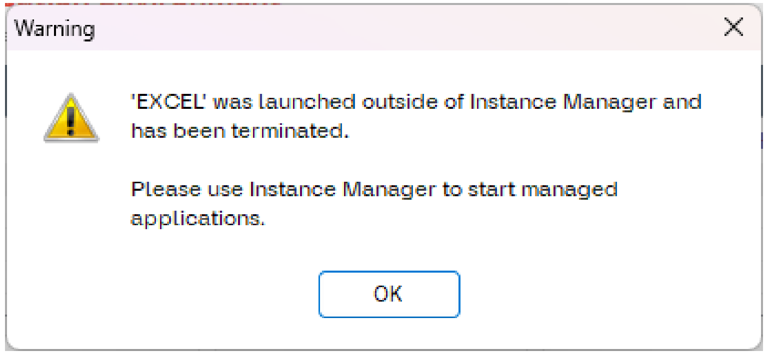


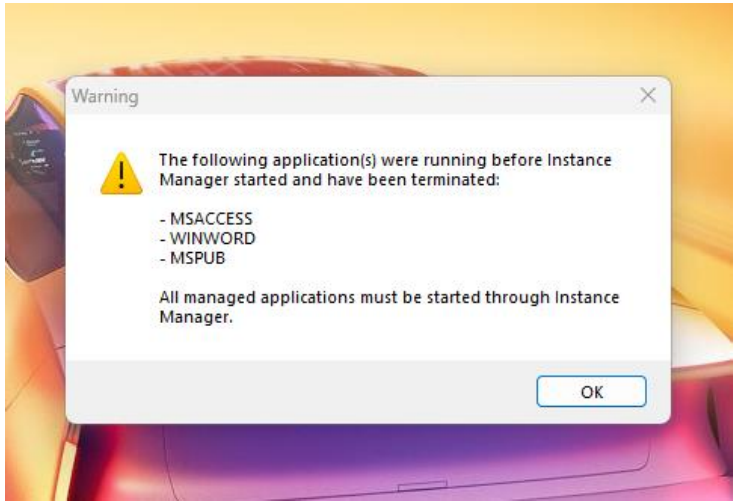


How It Works

– When you start an app through Instance Manager, it's added to an authorized list
– At each poll interval (default: 10 seconds, configurable in Settings), the watchdog checks all managed applications
– If a managed app is running but **not** in the authorized list, it was launched externally
– The unauthorized process is **killed immediately**
– A warning notification is shown (once per attempt)

Security
Managed applications can **only** be launched through its interface.

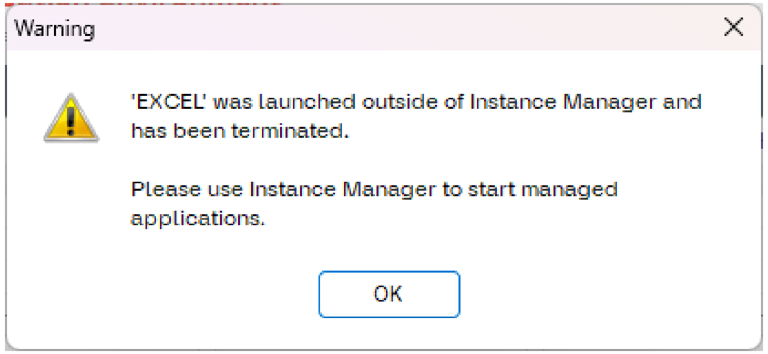


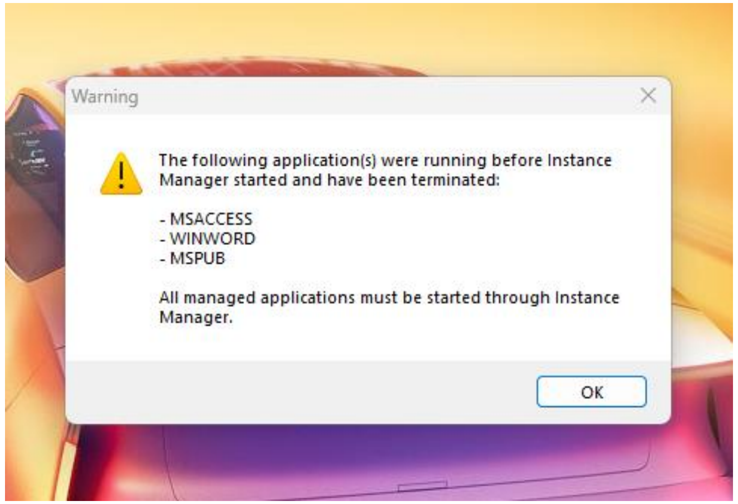


Scenarios

Scenario	What happens
User starts app via Instance Manager	Allowed – added to authorized list
User double-clicks the .exe directly	Killed within one poll interval + warning shown
User opens app via shortcut	Killed within one poll interval + warning shown
App was running before Instance Manager started	Terminated on startup + notification
App crashes and auto-restarts (Keep Open)	Allowed – re-authorized automatically
Same app running in another group	Shown as "Running (Other Group)" – must stop in the other group first

Security
Managed applications can **only** be launched through its interface.





Why this matters

– Ensures only one instance of each managed application runs
– Provides an audit trail of who started what and when
– Prevents circumventing instance management by launching apps directly

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FAQs

Complete Application User Guide

IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Frequently Asked Questions

Can I manage applications that require Administrator privileges?

– No. Instance Manager runs at standard user level and cannot manage elevated processes. If your managed application requires elevation, Instance Manager won't be able to start or stop it.

What happens if I close Instance Manager while applications are running?

- The managed applications **continue running** – Instance Manager does not stop them on exit. However, the watchdog will no longer monitor them. When you restart Instance Manager, any still-running managed apps will be terminated.

Can I manage the same application in multiple groups?

- Yes, you can add the same executable to different groups. Each entry is tracked independently. However, you cannot run the same application simultaneously from multiple groups. If you try to start an application that is already running in another group, Instance Manager will show a warning telling you which group its running in. You must stop it in the other group first before starting it in a new one. The status column will show Running (Other Group) in dark cyan for entries whose executable is running from a different group.

Frequently Asked Questions

Does Instance Manager modify my applications?

– No. Instance Manager only starts and stops processes. It never modifies any application files.

What if my application doesn't have a visible window?

– Instance Manager detects running applications by checking for a main window handle. Applications that run purely in the background (no window at all) may not be detected as "Running" and could be falsely treated as crashed.

What if my application shows an error dialog (unhandled exception)?

– If a Keep Open application becomes unresponsive due to an error dialog (e.g., an unhandled .NET exception dialog), Instance Manager will detect the unresponsive state after two consecutive poll cycles and force-kill the process. This allows auto-restart to take over without requiring manual intervention.

Can I change the watchdog polling interval?

–Yes. Click **Settings** and adjust the **Status Poll Interval** value. The default is 10000ms (10 seconds). Lower values make detection faster but use more CPU.

Frequently Asked Questions

What if my application shows an error dialog (unhandled exception)?
Instance Manager has multiple layers of detection for error dialogs:
1. **Error Dialog Pattern Matching** – If the dialog's window title contains a known error keyword (e.g., "unhandled exception", "NullReferenceException", "fatal error"), it is detected automatically and the process is force-killed after 2 consecutive poll cycles.
2. **Not Responding Detection** – If the error dialog blocks the UI thread and the window becomes unresponsive, it is detected via `Process.Responding` and force-killed after the configured timeout.
3. **Title Change Detection** – If the error dialog's title doesn't match any known pattern (e.g., a WinForms `ThreadExceptionDialog` that uses the app's product name as its title), enable **Detect Title Change** in the Edit dialog. The watchdog will detect the title change and force-kill the process.

All three mechanisms trigger auto-restart for Keep Open applications.

Frequently Asked Questions

How do I set a memory limit for my application?
1. Click **Edit** on the application
2. Set **Memory Limit (MB)** to your desired maximum (e.g., 500 for 500 MB)
3. Click OK
4. If the application's working set exceeds this limit, it will be force-killed and auto-restarted (if Keep Open is enabled)

Tip: Monitor your app's normal memory usage in Task Manager first, then set the limit to 2–3× the expected peak.

What is the "Last Exit Code" in the Edit dialog?
The Last Exit Code shows the exit code from the most recent process termination. A value of 0 typically means the process exited normally. A non-zero value (shown in red with "abnormal") usually indicates a crash or error.
Common non-zero codes:
-1073741819 (0xC0000005) – Access violation
-1073740791 (0xC0000409) – Stack buffer overrun
-532462766 (0xE0434352) – .NET unhandled exception

Frequently Asked Questions

Where is my data stored?
All data files are in the same folder as **IntelligentMutexExecutionEnvironment.exe** :
– `applications.json` – your managed applications
– `groups.json` – your groups
– `settings.json` – station name, performance, and logging configuration
– `logs/` – daily log files

How do I reset everything?
Delete `applications.json`, `groups.json`, and `settings.json`, then restart Instance Manager.

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Troubleshooting

Complete Application User Guide

IT/LMA – John Moises Paunlagui

February 18, 2026

Enabling Function

Troubleshooting Guide

Application Won't Start
1. Check that the file path is correct – click **Edit** and verify the Directory
2. Make sure the `.exe` file exists at that path
3. Check the log file for error messages
4. Verify you have permission to run the application

Application Won't Stop
1. Check the log file for "Access denied" errors
2. The application may have already closed externally
3. Click **Refresh** to force a status update
4. Try stopping it via Task Manager as a last resort

Status Stuck on "Starting..."
1. The application has been launched but its window hasn't appeared yet. Possible causes:
2. The application is still loading (wait for the grace period, default: 10 seconds, configurable in Settings)
3. The application runs without a visible window (not supported for status detection)
4. The application crashed immediately after launch (check logs)

Troubleshooting Guide

Status Shows "Failed"
The application has exhausted its maximum retry attempts. To fix:
1. Investigate why the application keeps crashing
2. Click **Edit** on the application
3. Reset the **Retry Count** to 0
4. Click OK
5. Click **Start** to try again

Application Keeps Getting Killed by Memory Limit
The memory limit may be set too low for the application's normal operation:
1. Click **Edit** on the application
2. Either increase the **Memory Limit (MB)** value or set it to `0` to disable
3. Click OK

Tip: Use Task Manager to observe the application's working set under normal conditions and set the limit well above that baseline.

Troubleshooting Guide

Application Keeps Getting Killed by Title Change Detection
The application legitimately changes its window title (e.g., showing document names, status text, or progress):
1. Click **Edit** on the application
2. Uncheck **Detect Title Change**
3. Click OK

Application Keeps Getting Killed by CPU-Hung Detection
1. The application legitimately uses sustained high CPU (e.g., rendering, video encoding):
CPU-hung detection is always active and cannot be disabled per-app. However, it requires **95% CPU usage across all cores for 3 consecutive poll cycles** (default: 30 seconds total). Most legitimate high-CPU operations won't trigger this unless they consume 100% of all CPU cores for an extended period.
2. If this is a problem, increase the **Status Poll Interval** in Settings to give the detection more time between checks.

Troubleshooting Guide

Unauthorized Launch Warning Keeps Appearing

This means someone (or something) keeps trying to launch the managed application outside of Instance Manager. Check for:

1. Scheduled tasks that launch the application
2. Startup items in Windows
3. Other users launching the application directly

Troubleshooting Guide

Logs Not Being Created

1. Make sure Instance Manager has write permission in its directory
2. Check if the `logs/` folder exists – if not, try creating it manually
3. Restart Instance Manager

Data File Corruption

If `applications.json` or `groups.json` becomes corrupted:

1. Check for a `.bak` or `.tmp` backup file in the same directory
2. Rename the `.tmp` file to replace the corrupted file
3. If no backup exists, delete the corrupted file (you'll lose that data)
4. Restart Instance Manager

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System Requirements

Complete Application User Guide

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February 18, 2026

Enabling Function

System Requirement
Minimum Requirement

Requirement	Minimum
Operating System	Windows XP SP3 or later
.NET Framework	4.0 or higher
RAM	50 MB available
Disk Space	5 MB + log files
Display	1024×768 or higher

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